Finanças II – Corporativas (MPE Insper)

Exercise list R3 — A frictionless benchmark, taxes, distress and static trade-off

Worked solutions

Statements are in `Listas/lista-r3-mm-taxes-tradeoff.md` and are not restated; each section cites the exercise by number and printed page. Editions cited are the local copies: Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt & Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001). Every number below is reproduced by `lista-r3-mm-taxes-tradeoff-solutions_check.py`. Amounts are in dollars unless a section says "millions". Each section follows the same order: what is asked, approach, notation, steps (with the answer to each part boxed), intuition, common mistakes, book answer and reference. Sections 16–18 solve *original* exercises on the bank capital debate (no course book has one); they have no book answer and cite Prof. Ricca's slides instead of a book page.

1. Two-outcome project financed with equity or safe debt — Berk 16.1, p. 490

What is asked. (a) The NPV of the project: what it is worth today minus what it costs. (b) How much cash the entrepreneur raises by selling the whole project to equity investors, i.e. the market value today of an all-equity claim on the project's cash flows. (c) If the \$100,000 is borrowed at the risk-free rate instead, what the shareholders receive in each state next year, and what that levered equity is worth today according to MM.

Approach. Part (a) is a plain NPV: discount the expected cash flow at the project's cost of capital (20%), which prices its risk. Part (b) uses the law of one price: unlevered equity receives exactly the project's cash flows, so it is worth exactly the project's PV. Part (c) is *MM Proposition I* in a perfect market (the section heading rules out taxes and distress costs): debt plus levered equity are claims to the same cash flows, so $E_L + D = V_U$. A last step checks the answer with *MM Proposition II*, which says how much more the riskier levered equity must earn.

Notation.

- $C_{\text{low}} = 130{,}000$, $C_{\text{high}} = 180{,}000$: project free cash flow at $t = 1$ in the two states; $p = \frac{1}{2}$: probability of the low state.
- $I = 100{,}000$: initial investment; $r_U = 0.20$: project (unlevered) cost of capital; $r_f = 0.10$: risk-free rate.
- V_U: value of the project's cash flows (= unlevered equity E_U); $NPV = V_U - I$.
- $D = 100{,}000$: amount borrowed at r_f; E_L: value of levered equity; r_E: its expected return; $r_D = r_f$.

Step 1 — Expected cash flow. The cost of capital is applied to the expected cash flow, so first average the two equally likely outcomes:

$$\begin{aligned} \mathbb{E}[C] &= p\,C_{\text{low}} + (1-p)\,C_{\text{high}} \\ &= (0.5)(130{,}000) + (0.5)(180{,}000) \\ &= 65{,}000 + 90{,}000 \\ &= 155{,}000. \end{aligned} \tag{1.1}$$

Step 2 — (a) Present value and NPV. Discount (1.1) one year at the project's cost of capital

and subtract the investment:

$$\begin{aligned} V_U &= \frac{\mathbb{E}[C]}{1+r_U} \qquad (1.2)\\ &= \frac{(155{,}000)}{(1.20)}\\ &= 129{,}166.67,\\ NPV &= V_U - I \qquad (1.3)\\ &= (129{,}166.67) - (100{,}000)\\ &= 29{,}166.67. \end{aligned}$$

Answer (a). $NPV = \$29{,}166.67$.

Step 3 — (b) Unlevered equity. All-equity holders receive exactly the project's cash flow in each state, so by the law of one price their claim is worth the PV in (1.2):

$$E_U = V_U = 129{,}166.67.$$

Selling the equity raises \$129,166.67, which exceeds I by the NPV: the entrepreneur keeps \$29,166.67 after paying for the investment.

Answer (b). The unlevered equity is worth (and raises) \$129,166.67.

Step 4 — (c) Repayment and the levered equity's cash flows. The loan is repaid with interest in one year:

$$\begin{aligned} D(1+r_f) &= (100{,}000)(1.10)\\ &= 110{,}000. \qquad (1.4) \end{aligned}$$

Since $110{,}000 < C_{\text{low}} = 130{,}000$, the firm repays in both states: the debt is indeed riskless, which justifies borrowing at r_f. Equity receives what is left:

$$\begin{aligned} \text{low state:} \quad & C_{\text{low}} - 110{,}000 = 130{,}000 - 110{,}000 = 20{,}000,\\ \text{high state:} \quad & C_{\text{high}} - 110{,}000 = 180{,}000 - 110{,}000 = 70{,}000. \end{aligned}$$

Step 5 — (c) Value by MM I. With no taxes or distress costs, MM I says the debt and the levered equity together are worth the unlevered firm, $E_L + D = V_U$. Solving for E_L:

$$\begin{aligned} E_L &= V_U - D \qquad (1.5)\\ &= (129{,}166.67) - (100{,}000)\\ &= 29{,}166.67. \end{aligned}$$

Step 6 — Consistency with MM II. The expected equity cash flow is $(0.5)(20{,}000) + (0.5)(70{,}000) = 10{,}000 + 35{,}000 = 45{,}000$, so the expected return implied by (1.5) is

$$\begin{aligned} r_E &= \frac{45{,}000}{E_L} - 1 \qquad (1.6)\\ &= \frac{45{,}000}{(29{,}166.67)} - 1\\ &= 0.542857. \end{aligned}$$

MM II without taxes, $r_E = r_U + \frac{D}{E_L}(r_U - r_D)$, needs the debt-equity ratio

$$\begin{aligned} \frac{D}{E_L} &= \frac{(100{,}000)}{(29{,}166.67)} = 3.428571,\\ r_E &= (0.20) + (3.428571)(0.20 - 0.10)\\ &= (0.20) + (3.428571)(0.10)\\ &= 0.542857, \end{aligned}$$

the same as (1.6).

Answer (c). Levered equity pays \$20,000 (low) or \$70,000 (high); by MM it is worth \$29,166.67 (expected return 54.29%).

Intuition. Borrowing slices the same cash flows differently; it does not change their total value. The entrepreneur captures the same NPV, \$29,166.67, either as the excess of the equity sale over I or as the value of the levered equity left after the loan. The levered equity bears all the project's risk on a smaller base, so it must offer a higher expected return (54.29% against 20%).

Common mistakes.

- Discounting the levered equity's expected cash flow at the project's 20%: $45{,}000/1.20 = 37{,}500$ overvalues it, because levered equity is riskier than the project.
- Subtracting only the principal (\$100,000) from the cash flows and forgetting the \$10,000 of interest.
- Not checking that the debt is riskless: the risk-free rate is the right borrowing rate only because \$110,000 is below the worst cash flow.

Book answer. None in the book.

Ref: Berk Ch. 16, §16.2 "Capital Structure in Perfect Capital Markets", pp. 464–472 (MM I, MM II).

2. One-time dividend or repurchase in a perfect market — Berk 17.7, p. 528

What is asked. (a) The share price right after the \$250 million is paid out as a dividend. (b) The share price after the same cash is used to buy back shares instead. (c) Which of the two policies leaves investors richer in a perfect capital market.

Approach. In a perfect market the share price equals the firm's market value per share, and that value includes the excess cash. Paying the cash out shrinks the firm by exactly the cash paid. With a dividend the number of shares is fixed, so the price falls; with a repurchase at a fair price the number of shares falls in proportion, so the price is unchanged. This is *MM payout irrelevance*: compare total wealth per share (stock plus cash), not prices.

Notation.

- $N = 500$ million shares; $P_0 = 15$: cum-payout share price; $C = 250$ million: excess cash paid out.
- d: dividend per share; P_{ex}: ex-dividend price; n_R: shares repurchased (millions); P_R: post-repurchase price.

Step 1 — Market value of the firm. With no debt, the firm's value is its equity, cash included:

$$NP_0 = (500)(15) = 7{,}500 \text{ million},$$

$$\text{of which operating assets} = 7{,}500 - 250 = 7{,}250 \text{ million}.$$

Step 2 — (a) Ex-dividend price. The dividend per share is the cash divided by the shares; after it is paid, the firm holds only the operating assets:

$$d = \frac{C}{N} = \frac{(250)}{(500)} = 0.50, \tag{2.1}$$

$$\begin{aligned} P_{\text{ex}} &= \frac{NP_0 - C}{N} \qquad (2.2)\\ &= \frac{(7{,}500) - (250)}{(500)}\\ &= \frac{7{,}250}{500}\\ &= 14.50. \end{aligned}$$

The price falls by exactly the dividend: $15 - 0.50 = 14.50$.

Answer (a). $P_{\text{ex}} = \$14.50$.

Step 3 — (b) Repurchase. In a perfect market the shares are bought at the fair cum-payout price P_0, so the cash buys C/P_0 shares and the operating assets are spread over the remaining shares:

$$n_R = \frac{C}{P_0} = \frac{(250)}{(15)} = 16.6667, \tag{2.3}$$

$$\begin{aligned} P_R &= \frac{NP_0 - C}{N - n_R} \qquad (2.4)\\ &= \frac{(7{,}500) - (250)}{(500) - (16.6667)} \\ &= \frac{7{,}250}{483.3333} \\ &= 15.00. \end{aligned}$$

Answer (b). $P_R = \$15.00$ (16.67 million shares repurchased).

Step 4 — (c) Comparison of wealth. Per original share, the dividend leaves stock plus cash of

$$P_{\text{ex}} + d = (14.50) + (0.50) = 15.00,$$

by (2.1)–(2.2); the repurchase leaves either a share worth 15.00 by (2.4) or 15.00 in cash for a tendered share. Wealth is \$15 per share under both.

Step 5 — Homemade dividend. An investor with 1,000 shares who prefers the dividend but gets the repurchase sells $500/15 = 33.33$ shares at \$15 to raise the \$500 the dividend would have paid, and keeps $1{,}000 - 33.33 = 966.67$ shares worth $(1{,}000 - 500/15)(15) = 15{,}000 - 500 = 14{,}500$: exactly the dividend outcome, \$14,500 in stock plus \$500 in cash.

Answer (c). Neither: wealth is \$15 per share either way, and each investor can convert one policy into the other at no cost.

Intuition. Paying out cash shrinks the firm by exactly the cash paid. With a dividend the price falls by the dividend; with a repurchase the share count falls instead and the price stays put. Any investor can create the other pattern for herself (sell shares, or reinvest the dividend), so payout policy is irrelevant (MM).

Common mistakes.

- Thinking the repurchase raises the price because there are fewer shares, forgetting that the \$250 million of cash leaves the firm too.
- Leaving the ex-dividend price at \$15: the share no longer carries the right to the cash.
- Comparing prices instead of wealth: $\$14.50 < \$15$ does not mean the dividend makes investors poorer, since they also hold \$0.50 of cash.

Book answer. None in the book.

Ref: Berk Ch. 17, §17.2 "Dividends Versus Share Repurchases in a Perfect Capital Market", pp. 503–508.

3. MM I with corporate taxes — Ross 16.14, p. 578

What is asked. The value of Frederico & Co. as it is (all equity), and its value if it borrows \$60,000 permanently and uses the money to buy back shares.

Approach. This is *MM Proposition I with corporate taxes*. The all-equity value is the after-tax perpetual cash flow capitalised at the unlevered cost of capital R_U. Perpetual debt creates a perpetual

interest tax shield $T_C R_D D$; it is as risky as the debt, so it is discounted at R_D and is worth $T_C D$. The levered value is $V_L = V_U + T_C D$; the buyback only changes who holds the claims.

Notation.

- $EBIT = 92{,}000$ per year forever; $R_U = 0.15$: unlevered cost of equity; $R_D = 0.09$: borrowing rate.
- $T_C = 0.35$: corporate tax rate; $D = 60{,}000$: perpetual debt used to buy back shares.
- V_U, V_L: unlevered and levered firm values; E: levered equity value; $PV(\text{TS})$: value of the tax shield.

Step 1 — After-tax cash flow of the all-equity firm. With no interest, the whole EBIT is taxed:

$$\begin{aligned} EBIT(1 - T_C) &= (92{,}000)(1 - 0.35) \\ &= (92{,}000)(0.65) \\ &= 59{,}800 \text{ per year forever.} \end{aligned}$$

Step 2 — Unlevered value. A level perpetuity is worth its cash flow divided by its discount rate:

$$\begin{aligned} V_U &= \frac{EBIT(1 - T_C)}{R_U} \\ &= \frac{(59{,}800)}{(0.15)} \\ &= 398{,}666.67. \end{aligned} \tag{3.1}$$

Step 3 — Value of the tax shield. The yearly interest is $R_D D = (0.09)(60{,}000) = 5{,}400$, and deducting it saves T_C times that in tax:

$$T_C R_D D = (0.35)(5{,}400) = 1{,}890 \text{ per year.}$$

This perpetuity is as risky as the debt, hence discounted at R_D; the R_D cancels:

$$\begin{aligned} PV(\text{TS}) &= \frac{T_C R_D D}{R_D} \\ &= T_C D \\ &= \frac{(1{,}890)}{(0.09)} = (0.35)(60{,}000) \\ &= 21{,}000. \end{aligned} \tag{3.2}$$

Step 4 — Levered value (MM I with taxes). Adding (3.2) to (3.1),

$$\begin{aligned} V_L &= V_U + T_C D \\ &= (398{,}666.67) + (21{,}000) \\ &= 419{,}666.67, \\ E &= V_L - D \\ &= (419{,}666.67) - (60{,}000) \\ &= 359{,}666.67. \end{aligned} \tag{3.3}$$

Answer (firm values). Unlevered: $V_U = \$398{,}666.67$. With \$60,000 of debt: $V_L = \$419{,}666.67$ (debt \$60,000, equity \$359,666.67).

Intuition. The government's claim on EBIT falls by \$1,890 a year once interest is deductible; that perpetual saving, worth $T_C D$, accrues to the firm's investors. The borrowing rate does not affect the answer because it scales both the saving and its discount rate.

Common mistakes.

- Capitalising pre-tax EBIT (92,000/0.15) instead of the after-tax cash flow.
- Discounting the tax shield at R_U: 1,890/0.15 = 12,600 understates it; with fixed perpetual debt the shield is as safe as the debt.
- Adding the whole \$60,000 to firm value: the debt replaces equity; only $T_C D$ is new value.

Book answer. None in the local copy (selected answers are in the web-only Apêndice C).

Ref: Ross Cap. 16, §16.4 "Proposições I e II de M&M com impostos sobre o lucro da pessoa jurídica", pp. 551–556.

4. APV with constant debt — Higgins 8.23, p. 333

What is asked. The adjusted present value of Gila's project: its base-case NPV plus the value of the interest tax shields from \$1 million of debt held for three years.

Approach. APV values the project as if all-equity financed (given: \$500,000) and then adds the PV of each financing side effect; here the only one is the *interest tax shield*. The debt is a fixed amount for three years, so the shield is a three-year annuity, and the statement says to discount it at the 8% debt rate.

Notation.

- $NPV_0 = 500{,}000$: base-case NPV of the project's free cash flows.
- $D = 1{,}000{,}000$: constant debt for $T = 3$ years; $r = 0.08$: interest and discount rate of the shields; $t = 0.40$: tax rate.
- TS: annual interest tax shield, received at the end of years 1–3; A: annuity factor; APV: adjusted present value.

Step 1 — Annual shield. Interest is charged on the constant balance, and the shield is the tax rate times the interest:

$$\begin{aligned} \text{interest} &= D\,r = (1{,}000{,}000)(0.08) = 80{,}000, \\ TS &= t \cdot D\,r \\ &= (0.40)(80{,}000) \\ &= 32{,}000. \end{aligned} \tag{4.1}$$

Step 2 — Three-year annuity factor at 8%. Discount each year's \$1 and add:

$$\begin{aligned} A &= \frac{1}{1.08} + \frac{1}{1.08^2} + \frac{1}{1.08^3} \\ &= (0.925926) + (0.857339) + (0.793832) \\ &= 2.577097. \end{aligned} \tag{4.2}$$

The closed form gives the same: $A = \left[1 - 1/1.08^3\right]/0.08 = \left[1 - 1/1.259712\right]/0.08 = 2.577097$.

Step 3 — PV of the shields and APV. Multiply the level shield (4.1) by (4.2) and add the base case:

$$\begin{aligned} PV(\text{TS}) &= TS \cdot A \\ &= (32{,}000)(2.577097) \\ &= 82{,}467.10, \end{aligned} \tag{4.3}$$

$$\begin{aligned} APV &= NPV_0 + PV(\text{TS}) \\ &= (500{,}000) + (82{,}467.10) \\ &= 582{,}467.10. \end{aligned} \tag{4.4}$$

Answer (APV). $APV = \$582{,}467.10$.

Intuition. APV separates the value of the operations ($500,000) from the value created by the financing side effect. With a fixed debt the shields are as safe as the debt, so they are discounted at the debt rate.

Common mistakes.

- Treating the shield as perpetual (tD = 400,000, APV $900,000): the debt lives only three years.
- Adding the interest ($80,000 a year) instead of the tax saved on it ($32,000).
- Discounting the shields at the cost of equity, which the statement rules out.

Book answer. Matches the book's answer, $582,467 (p. 425).

Ref: Higgins Ch. 8, Appendix "Asset Beta and Adjusted Present Value", pp. 318–325 (APV, p. 321).

5. MM II with a rising cost of debt — Berk 16.5, p. 491

What is asked. (a) Hardmon's expected equity return if it levers to $D/E = 0.50$ and pays 6% on debt. (b) The same with $D/E = 1.50$ and an 8% debt rate. (c) Whether shareholders are best served by the capital structure with the highest expected stock return.

Approach. With no taxes or distress costs (section heading), *MM I* fixes the firm's total value and *MM II* follows: the expected return on the assets (the WACC) stays at the unlevered 12%, so the cost of equity rises with leverage to keep the weighted average at 12%. We derive MM II from that average and apply it, letting r_D change with the debt level.

Notation.

- $r_U = 0.12$: expected return on unlevered equity (= on the assets, unchanged by leverage without taxes).
- D/E: debt-equity ratio (0.50 or 1.50); r_D: debt cost of capital (0.06 or 0.08); r_E: expected return on levered equity; $V = D + E$.

Step 1 — Derive MM II. The assets' return is the value-weighted average of the returns on the claims that finance them:

$$r_U = \frac{E}{D+E}\, r_E + \frac{D}{D+E}\, r_D.$$

Multiply both sides by $D + E$:

$$(D+E)\, r_U = E\, r_E + D\, r_D.$$

Move Dr_D to the left and write $(D+E)r_U = Er_U + Dr_U$:

$$E\, r_E = E\, r_U + D\, r_U - D\, r_D = E\, r_U + D\,(r_U - r_D).$$

Divide by $E > 0$:

$$r_E = r_U + \frac{D}{E}\,(r_U - r_D). \tag{5.1}$$

Step 2 — (a) $D/E = 0.50$, $r_D = 6\%$. Substituting into (5.1),

$$\begin{aligned} r_E &= (0.12) + (0.50)(0.12 - 0.06) \\ &= (0.12) + (0.50)(0.06) \\ &= (0.12) + (0.03) \\ &= 0.15. \end{aligned} \tag{5.2}$$

Answer (a). $r_E = 15\%$.

Step 3 — (b) $D/E = 1.50$, $r_D = 8\%$. Substituting into (5.1),

$$\begin{aligned} r_E &= (0.12) + (1.50)(0.12 - 0.08) \\ &= (0.12) + (1.50)(0.04) \\ &= (0.12) + (0.06) \\ &= 0.18. \end{aligned} \tag{5.3}$$

Answer (b). $r_E = 18\%$.

Step 4 — Check: the WACC is unchanged. Convert D/E into weights: $E/V = 1/(1 + D/E)$ and $D/V = (D/E)/(1 + D/E)$. In (a) $E/V = 1/1.5 = 0.666667$, $D/V = 0.5/1.5 = 0.333333$; in (b) $E/V = 1/2.5 = 0.4$, $D/V = 1.5/2.5 = 0.6$. Then

(a) $(0.666667)(0.15) + (0.333333)(0.06) = 0.10 + 0.02 = 0.12$,

(b) $(0.4)(0.18) + (0.6)(0.08) = 0.072 + 0.048 = 0.12$.

Step 5 — (c) Reply to the manager. The higher expected return in (5.3) is pure compensation for higher risk: equity bears the same business risk on a thinner base. By MM I the firm's value, and therefore the share price, is the same at every debt level; the WACC stays at 12% (Step 4). Choosing leverage to maximise r_E does not raise shareholder wealth; shareholders can also lever up on their own (homemade leverage) if they want a riskier, higher-return position.

Answer (c). Reject the argument: under MM a higher r_E only compensates for higher risk; firm value and the share price do not change, so shareholders are no better off.

Intuition. The cost of equity rises linearly with D/E at slope $r_U - r_D$. When debt becomes riskier (r_D rises), part of the risk moves to creditors and r_E rises less steeply, but the total required return on the assets is still r_U.

Common mistakes.

- Using D/V instead of D/E in MM II: in (a) that gives $0.12 + (0.333333)(0.06) = 0.14$.
- Keeping $r_D = 6\%$ in (b): $0.12 + (1.50)(0.06) = 0.21$ overstates r_E, since riskier debt takes on part of the risk.
- Equating a higher expected return with a higher value: value depends on cash flows and their risk together.

Book answer. None in the book.

Ref: Berk Ch. 16, §16.2, pp. 464–472 (MM II, leverage and the cost of equity).

6. Tax shields on amortising debt — Berk 16.11, p. 492

What is asked. The present value of the interest tax shields generated by a \$100 million loan at 10% that is paid down by \$25 million a year, with a 40% tax rate and shields as risky as the loan.

Approach. The *interest tax shield* each year is τ_c times that year's interest, and interest is charged on the balance still outstanding, so the shields shrink as the loan amortises. Because the debt is not permanent, the shortcut $\tau_c D$ does not apply: build the year-by-year schedule and discount each shield at the loan rate, as the statement prescribes.

Notation.

- B_t: debt outstanding during year $t = 1, \ldots, 4$ (millions); $B_1 = 100$, $B_{t+1} = B_t - 25$.
- $r_D = 0.10$: interest rate and discount rate of the shields; $\tau_c = 0.40$: marginal corporate tax rate.

- $TS_t = \tau_c r_D B_t$: tax shield received at the end of year t.

Assumption. The first \$25 million repayment is made one year from now, together with the first interest payment, and interest in year t is charged on the balance at the start of the year. The loan is then repaid after four years.

Step 1 — Schedule. Apply the 10% rate to each start-of-year balance and the 40% tax rate to each interest payment:

Year t	Balance B_t	Interest $r_D B_t$	Repayment	Shield $\tau_c r_D B_t$
1	100	10.0	25	4
2	75	7.5	25	3
3	50	5.0	25	2
4	25	2.5	25	1

$$TS_t = (0.40)(0.10)B_t = (0.04)B_t \in \{4,\ 3,\ 2,\ 1\}. \tag{6.1}$$

Step 2 — Discount factors at 10%. $1/1.10 = 0.909091$, $1/1.10^2 = 1/1.21 = 0.826446$, $1/1.10^3 = 1/1.331 = 0.751315$, $1/1.10^4 = 1/1.4641 = 0.683013$.

Step 3 — Present value. Multiply each shield (6.1) by its factor and add:

$$\begin{aligned} PV(\text{TS}) &= \sum_{t=1}^{4} \frac{TS_t}{(1+r_D)^t} \\ &= \frac{(4)}{1.10} + \frac{(3)}{1.21} + \frac{(2)}{1.331} + \frac{(1)}{1.4641} \\ &= 3.636364 + 2.479339 + 1.502630 + 0.683013 \\ &= 8.301346. \end{aligned} \tag{6.2}$$

Answer (PV of the shields). $PV(\text{TS}) \approx \$8.30$ million.

Intuition. Each year's shield is proportional to the debt still outstanding, so amortisation shrinks the shield stream. A permanent \$100 million loan would be worth $\tau_c D = \$40$ million in shields; repaying it over four years leaves $8.301346/40 = 0.2075$, about a fifth of that.

Common mistakes.

- Using $\tau_c D = \$40$ million, the value for *permanent* debt.
- Charging interest on the end-of-year balance (shields 3, 2, 1, 0), which gives $3/1.1 + 2/1.21 + 1/1.331 = 5.131480$ and misses the first year's interest.
- Adding the shields undiscounted (\$10 million) or discounting them at an unlevered rate.

Book answer. None in the book.

Ref: Berk Ch. 16, §16.3 "Debt and Taxes", pp. 473–479 (value of the interest tax shield).

7. Homemade leverage between twin firms — Ross 16.18, p. 578

What is asked. (a) An investor who can borrow at 6% wants 5% of Cavaleiro (levered). Can she earn more by buying 5% of Weber (unlevered) and borrowing personally, spending the same amount of her own money? (b) Which strategy investors will choose, and when this process stops.

Approach. This is *homemade leverage*, the arbitrage behind *MM Proposition I* without taxes. Two firms with the same operating income must have the same total value; if not, investors replicate one firm's equity with the other firm's equity plus personal borrowing or lending. Here $V_C = 1{,}632{,}000 + 1{,}200{,}000 = 2{,}832{,}000 > V_W = 2{,}500{,}000$, so Cavaleiro is overpriced and the cheaper route to its payoff goes through Weber.

Notation.

- X: perpetual operating income of each firm (expected 400,000; random, identical for both); no taxes.
- Weber (unlevered): $E_W = V_W = 2{,}500{,}000$. Cavaleiro (levered): $E_C = 1{,}632{,}000$, $D_C = 1{,}200{,}000$, interest $I_C = 72{,}000$ per year, so $r_D = 72{,}000/1{,}200{,}000 = 0.06$.
- $\alpha = 0.05$: fraction bought; $r = 0.06$: personal borrowing rate; L: amount borrowed personally.

Assumption. "Os custos líquidos das estratégias sejam iguais" is read as: both strategies require the same own funds today, the \$81,600 that 5% of Cavaleiro costs.

Step 1 — Strategy C: 5% of Cavaleiro. Cavaleiro's shareholders receive operating income minus interest, $X - I_C$; with the projected income that is $400{,}000 - 72{,}000 = 328{,}000$. Then

$$\text{cost}_C = \alpha E_C = (0.05)(1{,}632{,}000) = 81{,}600, \tag{7.1}$$

$$\text{income}_C = \alpha(X - I_C) = (0.05)(328{,}000) = 16{,}400, \tag{7.2}$$

a return of $16{,}400/81{,}600 = 20.10\%$.

Step 2 — Strategy W: 5% of Weber with the same own funds. The Weber stake costs $(0.05)(2{,}500{,}000) = 125{,}000$ and earns $(0.05)(400{,}000) = 20{,}000$. With own funds (7.1), the rest is borrowed at 6%:

$$L = 125{,}000 - 81{,}600 = 43{,}400, \tag{7.3}$$

$$\begin{aligned} \text{income}_W &= 20{,}000 - rL \\ &= (20{,}000) - (0.06)(43{,}400) \\ &= (20{,}000) - (2{,}604) = 17{,}396, \end{aligned} \tag{7.4}$$

a return of $17{,}396/81{,}600 = 21.32\%$.

Step 3 — (a) Same risk, higher income. Write both incomes for any realisation of X:

$$\begin{aligned} \text{income}_C &= 0.05\,(X - 72{,}000) = 0.05X - 3{,}600, \\ \text{income}_W &= 0.05X - (0.06)(43{,}400) = 0.05X - 2{,}604. \end{aligned}$$

Both move one-for-one with $0.05X$, so they carry the same operating risk, and W pays $3{,}600 - 2{,}604 = 996$ more in every state. The W position even uses less leverage: \$43,400 of personal debt against her 5% share of Cavaleiro's \$1,200,000, i.e. \$60,000.

Answer (a). Yes: for the same \$81,600, 5% of Weber plus \$43,400 borrowed pays \$17,396 (21.32%) against \$16,400 (20.10%), \$996 more in every state at no extra risk.

Step 4 — The underlying arbitrage. Replicating strategy C exactly needs the investor's share of Cavaleiro's debt, $L = \alpha D_C = (0.05)(1{,}200{,}000) = 60{,}000$: income $20{,}000 - (0.06)(60{,}000) = 20{,}000 - 3{,}600 = 16{,}400$, identical to (7.2), for own funds $125{,}000 - 60{,}000 = 65{,}000$. The saving is

$$81{,}600 - 65{,}000 = 16{,}600 = \alpha(V_C - V_W) = (0.05)(2{,}832{,}000 - 2{,}500{,}000) = (0.05)(332{,}000), \tag{7.5}$$

and investing it at 6% yields $(0.06)(16{,}600) = 996$ a year, the gap found in Step 3.

Step 5 — (b) What investors do and when it stops. Every Cavaleiro shareholder gains by selling Cavaleiro shares and buying Weber shares with personal borrowing. Selling pressure lowers E_C and buying pressure raises E_W until the arbitrage profit (7.5) is zero, i.e. until $V_C = V_W$ (MM I). If Weber's value stays at \$2.5 million, it stops when $E_C = 2{,}500{,}000 - 1{,}200{,}000 = 1{,}300{,}000$. Cavaleiro's cost of equity is then $328{,}000/1{,}300{,}000 = 25.23\%$, which is MM II with $r_U = 400{,}000/2{,}500{,}000 = 0.16$:

$$r_E = 0.16 + \tfrac{1.2}{1.3}(0.16 - 0.06) = 0.16 + (0.923077)(0.10) = 0.16 + 0.092308 = 0.252308.$$

Answer (b). Investors sell Cavaleiro and buy Weber on personal margin; the process ends when the two firms have the same total value, $V_C = V_W$ (e.g. $E_C = \$1{,}300{,}000$ if V_W stays at \$2.5 million).

Intuition. Investors can borrow on the same terms as the firm, so corporate leverage adds nothing they cannot build themselves. A levered firm that trades at a premium is selling something investors can make more cheaply at home, and arbitrage removes the premium.

Common mistakes.

- Comparing the firms' returns on equity without equalising own funds (or risk): Weber's unlevered 16% is not comparable with Cavaleiro's 20.10%.
- Borrowing the full \$60,000 and then comparing incomes: that equalises income, not cost; the gain then shows up as \$16,600 saved today instead of \$996 a year.
- Forgetting that the process stops at equal *firm* values, not equal share prices or equal returns.

Book answer. None in the local copy (selected answers are in the web-only Apêndice C).

Ref: Ross Cap. 16, §16.2 "O efeito da alavancagem financeira" (homemade leverage), pp. 542–547; §16.3, pp. 547–551 (Propositions I and II).

8. Announcement effect of a debt-financed repurchase with distress costs — Berk 16.22, p. 494

What is asked. (a) The share price after Hawar announces a permanent \$20 million debt-financed buyback, if corporate taxes are the only imperfection. (b) If the price actually rises only to \$5.75, the PV of the financial distress costs the market attributes to the new debt.

Approach. Permanent debt creates an interest tax shield worth $\tau_c D$ (*MM I with taxes*). When the plan is announced, the whole change in firm value is capitalised into the price of the existing shares, before any shares are bought back. With distress costs, the *static trade-off* identity $V_L = V_U + \tau_c D - PV(\text{FDC})$ lets us back out $PV(\text{FDC})$ from the observed price.

Notation.

- $P_0 = 5.50$, $N = 10$ million shares, so $E_0 = (5.50)(10) = 55$ million.
- $D = 20$ million of permanent new debt; $\tau_c = 0.30$; P_1: price after the announcement; $PV(\text{FDC})$: PV of financial distress costs.

Assumption. The announcement is credible and all the value change is priced at once; shares are later bought back at the post-announcement price, so the repurchase itself transfers no value.

Step 1 — (a) Value of the tax shield. Permanent debt adds (cf. (3.2))

$$\tau_c D = (0.30)(20) = 6 \text{ million.}$$

Step 2 — (a) Price after the announcement. The \$6 million goes to the 10 million shares outstanding at the announcement:

$$\begin{aligned} P_1 &= \frac{E_0 + \tau_c D}{N} \\ &= \frac{(55) + (6)}{(10)} \\ &= 6.10. \end{aligned} \tag{8.1}$$

Check that the buyback leaves it there: \$20 million buys $20/6.10 = 3.278689$ million shares; equity falls to $61 - 20 = 41$ million over $10 - 3.278689 = 6.721311$ million shares, i.e. $41/6.721311 = 6.10$.

Answer (a). $P_1 = \$6.10$ (a gain of \$0.60 per share).

Step 3 — (b) Distress costs implied by \$5.75. The observed gain in equity value is

$$(5.75)(10) - (55) = 57.5 - 55 = 2.5 \text{ million.}$$

By the trade-off identity this gain equals the tax benefit net of distress costs; solving for the distress costs:

$$\begin{aligned} 2.5 &= \tau_c D - PV(\text{FDC}) \\ PV(\text{FDC}) &= \tau_c D - 2.5 \\ &= (6) - (2.5) \\ &= 3.5 \text{ million.} \end{aligned} \tag{8.2}$$

(The buyback at \$5.75 again leaves the price unchanged: $20/5.75 = 3.478261$ million shares, $37.5/(10 - 3.478261) = 37.5/6.521739 = 5.75$.)

Answer (b). $PV(\text{FDC}) = \$3.5$ million.

Intuition. The market pays the shareholders for the expected tax savings and charges them for expected distress costs. The smaller price jump (\$0.25 against \$0.60) shows that the market expects distress costs to eat more than half of the tax benefit of this debt.

Common mistakes.

- Dividing the gain by the post-repurchase share count: the announcement gain accrues to all 10 million shares outstanding when the news arrives.
- Looking for an interest rate: with permanent debt the shield is worth $\tau_c D$ whatever the rate.
- Reading \$5.75 as a loss because it is below \$6.10: the \$3.5 million is the gap between the two prices times 10 million shares.

Book answer. None in the book.

Ref: Berk Ch. 16, §16.3 pp. 473–479 and §16.5 "Optimal Capital Structure: The Tradeoff Theory", pp. 480–482.

9. After-tax WACC with riskless perpetual debt, then a debt-for-equity exchange — Grinblatt & Titman 13.15–13.16, p. 494

What is asked. 13.15: Akron's expected equity return (from the CAPM) and its after-tax WACC with \$50 million of riskless perpetual debt and a 40% tax rate. 13.16: Akron's WACC after it retires half its debt by issuing \$25 million of equity to the debt holders, under the Hamada model's debt policy.

Approach. The CAPM gives r_E from β_E; G&T's (13.8) then gives the after-tax WACC. For 13.16 the *Hamada model* (static perpetual riskless debt) says the firm is unlevered assets plus a riskless tax shield $T_c D$. The unlevered assets, their beta and their return r_{UA} do not change with leverage, so we unlever once and then use the *MM adjusted cost of capital* $WACC = r_{UA}(1 - T_c D/V)$ at the new debt level, cross-checking through the relevered cost of equity.

Notation.

- $D = 50$, $E = 50$ (millions): perpetual riskless debt and equity; $A = D + E = 100$: market value of the assets.
- $\beta_E = 1.2$; $r_f = 0.08$; $\mathbb{E}[r_M] = 0.13$; CAPM holds; $r_D = r_f$ (riskless debt); $T_c = 0.40$.
- UA, TX: unlevered assets and debt tax shield, $A = UA + TX$; β_{UA}, r_{UA}: their beta and expected return.
- Primes (D', E', V', $WACC'$) denote values after the exchange offer.

Step 1 — 13.15: cost of equity (CAPM). The market risk premium is $0.13 - 0.08 = 0.05$, so

$$\begin{aligned} r_E &= r_f + \beta_E(\mathbb{E}[r_M] - r_f) \\ &= (0.08) + (1.2)(0.13 - 0.08) \\ &= (0.08) + (1.2)(0.05) \\ &= (0.08) + (0.06) \\ &= 0.14. \end{aligned} \tag{9.1}$$

Step 2 — 13.15: WACC. Use G&T's after-tax WACC (13.8) with market weights $50/100 = 0.5$ each and (9.1):

$$\begin{aligned} WACC &= \frac{E}{D+E}\, r_E + \frac{D}{D+E}\, r_D(1 - T_c) \\ &= (0.5)(0.14) + (0.5)(0.08)(1 - 0.40) \\ &= (0.07) + (0.5)(0.08)(0.60) \\ &= (0.07) + (0.024) \\ &= 0.094. \end{aligned} \tag{9.2}$$

Answer (13.15). $r_E = 14\%$; $WACC = 9.40\%$.

Step 3 — Unlevered cost of capital (Hamada). With static perpetual riskless debt, $TX = T_c D$ (13.2) and $UA = D + E - T_c D = 100 - (0.40)(50) = 100 - 20 = 80$ (13.3). Invert (13.6), $\beta_E = [1 + (1 - T_c)\frac{D}{E}]\beta_{UA}$, by dividing both sides by the bracket:

$$\begin{aligned} \beta_{UA} &= \frac{\beta_E}{1 + (1 - T_c)D/E} \\ &= \frac{(1.2)}{1 + (1 - 0.40)(50/50)} \\ &= \frac{1.2}{1 + 0.6} = \frac{1.2}{1.6} \\ &= 0.75, \end{aligned} \tag{9.3}$$

$$\begin{aligned} r_{UA} &= r_f + \beta_{UA}(\mathbb{E}[r_M] - r_f) \\ &= (0.08) + (0.75)(0.05) \\ &= (0.08) + (0.0375) \\ &= 0.1175. \end{aligned} \tag{9.4}$$

Step 4 — The adjusted cost of capital formula. Start from (13.11), $WACC = \bar{r}_A - \frac{D}{V} T_c r_D$, and substitute the Hamada form of (13.1b), $\bar{r}_A = (1 - \frac{T_c D}{V}) r_{UA} + \frac{T_c D}{V} r_D$:

$$\begin{aligned} WACC &= \Big(1 - \frac{T_c D}{V}\Big) r_{UA} + \frac{T_c D}{V} r_D - \frac{D}{V} T_c r_D \\ &= \Big(1 - \frac{T_c D}{V}\Big) r_{UA} \\ &= r_{UA}\Big(1 - T_c \frac{D}{V}\Big), \end{aligned} \tag{9.5}$$

since the two r_D terms are equal and cancel. Check at today's leverage: $(0.1175)(1 - (0.40)(0.5)) = (0.1175)(0.8) = 0.094$, equal to (9.2).

Step 5 — 13.16: values after retiring \$25 million of debt. The unlevered assets do not change; the shield shrinks to $T_c D' = (0.40)(25) = 10$. Hence

$$\begin{aligned} V' &= UA + T_c D' = (80) + (10) = 90, \\ E' &= V' - D' = (90) - (25) = 65, \end{aligned} \tag{9.6}$$

of which \$25 million is the new shares given to the old creditors and $65 - 25 = 50 - (0.40)(25) = 40$ is the old shareholders' stake: they bear the \$10 million loss of shield value.

Step 6 — 13.16: new WACC. The new debt ratio is $D'/V' = 25/90 = 0.277778$; substitute it into (9.5) with (9.4):

$$\begin{aligned} WACC' &= r_{UA}\Big(1 - T_c\frac{D'}{V'}\Big) \\ &= (0.1175)(1 - (0.40)(0.277778)) \\ &= (0.1175)(0.888889) \\ &= 0.104444. \end{aligned} \tag{9.7}$$

Step 7 — Cross-check through the cost of equity. Relever with (13.6) at $D'/E' = 25/65 = 0.384615$:

$$\begin{aligned} \beta'_E &= \beta_{UA}\Big[1 + (1 - T_c)\frac{D'}{E'}\Big] \\ &= (0.75)\big[1 + (0.6)(0.384615)\big] = (0.75)(1 + 0.230769) \\ &= (0.75)(1.230769) = 0.923077, \\ r'_E &= (0.08) + (0.923077)(0.05) = (0.08) + (0.046154) = 0.126154, \\ WACC' &= \tfrac{65}{90}(0.126154) + \tfrac{25}{90}(0.08)(0.6) \\ &= (0.091111) + (0.013333) = 0.104444, \end{aligned}$$

the same as (9.7). The WACC rises from 9.40% to 10.44%.

Answer (13.16). $WACC' = 10.44\%$.

Assumption. The debt ratio in (9.7) is measured at post-exchange market values. Using $D'/V = 25/100$ (ignoring the lost shield) would give $(0.1175)(1 - 0.40 \cdot 0.25) = 10.575\%$; that mixes the old firm value with the new debt and is not consistent with the Hamada model.

Intuition. In the Hamada model the only thing debt adds is a riskless tax shield. Cutting the debt removes part of that cheap, government-funded slice of the financing, so the WACC moves back toward the unlevered cost of capital, 11.75%.

Common mistakes.

- Computing the pre-tax WACC, $(0.5)(0.14) + (0.5)(0.08) = 0.11$, instead of (13.8) with $(1 - T_c)$ on debt.
- Keeping $r_E = 14\%$ after the exchange: $\frac{65}{90}(0.14) + \frac{25}{90}(0.048) = 0.114444$ ignores that less leverage makes equity safer.
- Weighting by the old value $V = 100$ (10.575%): retiring debt destroys \$10 million of shield, so $V' = 90$.

Book answer. None in the book.

Ref: Grinblatt & Titman Ch. 13: Hamada model, eqs. (13.1)–(13.7), pp. 463–466; WACC (13.8), p. 476; adjusted cost of capital formula, pp. 481–482.

10. Taxes vs. bankruptcy costs in two states — Grinblatt & Titman 16.2, p. 591

What is asked. (a) The value of Nigel's firm if all equity financed. (b) What the firm raises today by selling a zero-coupon bond that promises \$44 next period. (c) The value of the equity with that bond outstanding. (d) The total value of the firm with that bond. (e) Whether the firm would be worth more with a \$70 obligation instead.

Approach. Investors are risk neutral, so every claim is worth its expected payoff discounted at the 10% riskless rate. Debt brings a tax saving of \$0.05 per dollar raised; debt large enough to default in the bad state also triggers a \$20 bankruptcy cost. This is the *static trade-off*: firm value $= V_U + PV(\text{tax saving}) - PV(\text{bankruptcy costs})$. For each debt level we first check whether default happens, then price debt and equity state by state.

Notation.

- $C_G = 210$, $C_B = 66$: pre-tax cash flows next period in the favourable and unfavourable state, each with probability $\frac{1}{2}$.
- $t = \frac{1}{3}$: tax rate of the all-equity firm; $s = 0.05$: tax saved per dollar of debt financing; $r_f = 0.10$; investors are risk neutral.
- F: face value of the zero-coupon debt; D: its market value today; $K = 20$: bankruptcy cost; V_U, E, V: unlevered value, equity, levered firm value.

Assumption. "Debt financing used" is the amount raised today, D; the saving sD is realised next period in both states (if the firm defaults it accrues to the creditors who take it over). The answer to (e) does not depend on this reading (Step 7).

Step 1 — (a) All-equity value. After paying one third in tax, the firm keeps two thirds: $210(1 - \frac{1}{3}) = 140$ and $66(1 - \frac{1}{3}) = 44$. Discount the expected value:

$$\begin{aligned} V_U &= \frac{\frac{1}{2}(140) + \frac{1}{2}(44)}{1.10} \\ &= \frac{70 + 22}{1.10} = \frac{92}{1.10} = \frac{920}{11} \\ &= 83.6364. \end{aligned} \tag{10.1}$$

Answer (a). $V_U = \$83.64$.

Step 2 — (b) Debt with $F = 44$. Guess that the debt is safe; then it is worth its face discounted at the riskless rate, and the tax saving follows:

$$D = \frac{44}{1.10} = 40, \qquad sD = (0.05)(40) = 2.$$

Verify the guess: in the bad state the firm has $44 + 2 = 46 \geq 44$, so it pays in full in both states.

Answer (b). The firm receives $D = \$40$.

Step 3 — (c) Equity. Equity gets what is left after paying 44: $140 + 2 - 44 = 98$ in the good state and $46 - 44 = 2$ in the bad state:

$$\begin{aligned} E &= \frac{\frac{1}{2}(98) + \frac{1}{2}(2)}{1.10} \\ &= \frac{49 + 1}{1.10} = \frac{50}{1.10} = \frac{500}{11} \\ &= 45.4545. \end{aligned} \tag{10.2}$$

Answer (c). $E = \$45.45$.

Step 4 — (d) Firm value. Add the two claims:

$$\begin{aligned} V &= D + E \\ &= (40) + (45.4545) \\ &= 85.4545, \end{aligned} \tag{10.3}$$

equal to $V_U + \frac{2}{1.10} = \frac{920}{11} + \frac{20}{11} = \frac{940}{11}$: the PV of the tax saving, 1.8182, is the whole gain.

Answer (d). $V = \$85.45$.

Step 5 — (e) Debt with $F = 70$: price the risky debt. Now the bad-state cash, $44 + sD$, is below 70 for any plausible D, so the firm defaults; creditors take over, pay the \$20 cost and receive $44 + 0.05D - 20 = 24 + 0.05D$. In the good state they get 70. The debt's price solves

$$\begin{aligned} D &= \frac{\frac{1}{2}(70) + \frac{1}{2}(24 + 0.05D)}{1.10} \\ 1.10\,D &= 35 + 12 + 0.025\,D \\ 1.10\,D - 0.025\,D &= 47 \\ 1.075\,D &= 47 \\ D &= \frac{47}{1.075} = 43.72093. \end{aligned} \tag{10.4}$$

Then $sD = (0.05)(43.72093) = 2.18605$ and the bad-state recovery is $24 + 2.18605 = 26.18605 < 70$: default confirmed.

Step 6 — (e) Equity and firm value. Equity gets $140 + 2.18605 - 70 = 72.18605$ in the good state and 0 in the bad state:

$$\begin{aligned} E &= \frac{\frac{1}{2}(72.18605) + \frac{1}{2}(0)}{1.10} \\ &= 32.81184, \\ V &= D + E \\ &= (43.72093) + (32.81184) \\ &= 76.53277. \end{aligned} \tag{10.5}$$

Equivalently $V = V_U + PV(sD) - PV(K) = (83.63636) + (1.98732) - (9.09091) = 76.53277$, using $2.18605/1.10 = 1.98732$ and $\frac{1}{2}(20)/1.10 = 9.09091$.

Step 7 — Robustness. Whatever the basis of the saving, it is at most 0.05×70 next period, worth 3.18182, while expected bankruptcy costs are worth 9.09091; so $V \le (83.63636) + (3.18182) - (9.09091) = 77.72727 < 85.4545$ under any reading.

Answer (e). No: with a \$70 obligation the firm is worth \$76.53, less than \$85.45.

Intuition. This is the static trade-off in miniature. Safe debt adds a small tax subsidy at no cost. Debt large enough to default in the bad state buys a little more subsidy but triggers a deadweight loss of \$20 with probability one half, which dominates.

Common mistakes.

- Ignoring corporate tax in (a): $(105 + 33)/1.10 = 125.4545$ is the pre-tax value.
- Pricing the \$70 bond as safe, $70/1.10 = 63.64$: the bad state cannot pay it.
- Thinking the bankruptcy cost hurts only creditors: they price it into D, so the shareholders who issue the debt bear it.

Book answer. None in the book.

Ref: Grinblatt & Titman Ch. 16, §16.1 "Bankruptcy" (direct bankruptcy costs vs. the debt tax advantage), pp. 559–561.

11. Risky debt in a binomial market and new debt of equal seniority — Grinblatt & Titman 14.2–14.3, pp. 526–527

What is asked. 14.2: The share price of a firm worth \$210 or \$80 next period with \$50 of debt and 100 shares. 14.3: After a surprise issue of \$50 more face of equal-seniority debt used to buy back shares: (a) the price of the new debt; (b) the share price right after the announcement; (c) show that a 20% shareholder who undoes the leverage change ends up better off; (d) show that MM still holds.

Approach. With a zero interest rate and equal risk-neutral probabilities, every claim is worth the plain average of its two payoffs. Equal seniority means that in default the cash is shared pro rata by face value, so the new issue makes the old debt risky. *MM Proposition I* says total firm value does not depend on the split; any gain to shareholders must therefore be a transfer from the old creditors. Undoing leverage is *homemade leverage* run in reverse: hold debt alongside the shares.

Notation.

- $C_G = 210$, $C_B = 80$: firm cash flow in the good/bad state; risk-neutral probability $\frac{1}{2}$ each; $r_f = 0$; no taxes.
- $F_0 = 50$: face of the existing debt; $F_1 = 50$: face of the new equal-seniority debt; 100 shares initially.
- V: firm (machine) value; D_0, D_1: market values of old and new debt; P: share price; n_R: shares repurchased.

Step 1 — 14.2: share price. With $r_f = 0$, values are risk-neutral expectations:

$$V = \tfrac{1}{2}(210) + \tfrac{1}{2}(80) = 105 + 40 = 145. \tag{11.1}$$

The old debt is safe ($80 \geq 50$), so $D_0 = 50$; equity pays $(210 - 50,\ 80 - 50) = (160,\ 30)$ and is worth $145 - 50 = 95$, so $P = 95/100 = 0.95$.

Answer (14.2). $P = \$0.95$.

Step 2 — (a) Price of the new debt. Total face becomes $F_0 + F_1 = 100 > 80$, so in the bad state the \$80 is shared pro rata: each issue gets $\frac{50}{100}(80) = 40$. The new debt pays $(50,\ 40)$:

$$D_1 = \tfrac{1}{2}(50) + \tfrac{1}{2}(40) = 25 + 20 = 45. \tag{11.2}$$

The old debt now has the same payoff, so D_0 also falls from 50 to 45.

Answer (a). $D_1 = \$45$.

Step 3 — (b) Share price at the announcement. Before the buyback the firm holds the machine plus the \$45 cash raised and owes debt worth $2 \times 45 = 90$; the 100 shares are still outstanding:

$$E = (145) + (45) - (90) = 100, \qquad P = \frac{100}{100} = 1.00. \tag{11.3}$$

The firm then buys back $n_R = 45/1.00 = 45$ shares; 55 remain. Equity pays $(210 - 100,\ 0) = (110,\ 0)$, worth $\frac{1}{2}(110) = 55 = 145 - 90$, i.e. still \$1.00 per share.

Answer (b). The share price jumps from \$0.95 to \$1.00.

Assumption. "A shareholder with 20 percent of the shares" holds 20 of the 100 shares before the announcement; "undoing the leverage change" means rebuilding the pre-announcement payoff exposure with personal trades.

Step 4 — (c) Undoing the leverage. Before, she held 20% of an equity paying $(160, 30)$, i.e. $(32, 6)$, worth $\frac{1}{2}(32) + \frac{1}{2}(6) = 16 + 3 = 19$. After the announcement her 20 shares are worth 20. One way to undo the change: keep $0.2 \times 55 = 11$ shares (paying $11 \times (2, 0) = (22, 0)$), sell $20 - 11 = 9$ shares at \$1.00, and with the \$9 buy 20% of the new debt, $0.2 \times 45 = 9$, which pays $0.2 \times (50, 40) = (10, 8)$. The position pays

$$(22 + 10,\ 0 + 8) = (32,\ 8), \qquad \text{worth } \tfrac{1}{2}(32) + \tfrac{1}{2}(8) = 20. \tag{11.4}$$

This beats the old payoff $(32, 6)$ state by state; the gain $20 - 19 = 1$ is 20% of the \$5 lost by the old creditors ($0.2 \times 5 = 1$).

To rebuild $(32, 6)$ exactly, hold n shares (each paying $(2, 0)$) and a fraction β of all the debt (paying $\beta(100, 80)$). The bad state gives $80\beta = 6$, so $\beta = 0.075$; the good state gives $2n + (0.075)(100) = 32$, so $2n = 24.5$ and $n = 12.25$. The cost is $12.25(1.00) + 0.075(90) = 12.25 + 6.75 = 19.00$, and she keeps \$1 in cash from her \$20 of shares.

Answer (c). She can rebuild her old payoff $(32, 6)$ for \$19 while her shares are worth \$20: she is \$1 better off.

Step 5 — (d) MM still holds. Total firm value is $50 + 95 = 145$ before and $90 + 55 = 145$ after, as (11.1) requires: changing the capital structure did not change the value of the claims taken together, and investors can undo it (Step 4). The share price rose only because equal-seniority debt diluted the old creditors, a transfer of \$5 from them to the shareholders, not value creation.

Answer (d). Firm value is \$145 before and after; the \$1.00 price reflects a \$5 transfer from old creditors, not new value.

Intuition. MM is a statement about the size of the pie, not about how it is cut. Issuing new debt that ranks equally with old debt cuts the old creditors' slice; without protective covenants, shareholders pocket that slice.

Common mistakes.

- Pricing the new debt at its face \$50, ignoring that equal seniority makes all the debt risky.
- Dividing post-repurchase equity by 100 shares (or announcement equity by 55): the announcement price uses 100 shares and the cash still in the firm.
- Concluding that MM fails because the share price rose: the old creditors lost exactly what shareholders gained.

Book answer. None in the book.

Ref: Grinblatt & Titman Ch. 14, §14.1 "The Modigliani-Miller Theorem" p. 501, §14.2 "How an Individual Investor Can 'Undo' a Firm's Capital Structure Choice" p. 505, §14.3 "How Risky Debt Affects the Modigliani-Miller Theorem" pp. 506–508.

12. Leveraged recapitalisation with corporate taxes — Berk 16.18, p. 493

What is asked. (a) The market value of Kurz's assets before the announcement. (b) The market value of its assets, tax shield included, after the \$50 million is borrowed but before any shares are bought back. (c) The share price just before the buyback, and how many shares the \$50 million buys. (d) The market-value balance sheet and share price after the buyback.

Approach. Build *market-value balance sheets* at each date. Permanent interest-only debt creates an interest tax shield worth $\tau_c D$ (*MM I with taxes*), which is capitalised into the share price as soon as the plan is known. The buyback then happens at that fair price, so it moves no value between selling and remaining shareholders.

Notation.

- $N_0 = 20$ million shares at $P_0 = 7.50$; all equity before the announcement.
- $D = 50$ million of permanent interest-only debt; $\tau_c = 0.40$; A_1: assets after the issue; P_1: share price after the debt is issued; n_R: shares repurchased (millions).

Step 1 — (a) Existing assets. With no debt, assets equal equity:

$$N_0 P_0 = (20)(7.50) = 150 \text{ million}.$$

Answer (a). \$150 million.

Step 2 — (b) Assets after the issue, before the repurchase. The firm holds the old assets, the \$50 million cash, and the tax shield of permanent debt, $\tau_c D = (0.40)(50) = 20$:

$$A_1 = (150) + (50) + (20) = 220 \text{ million}. \tag{12.1}$$

Answer (b). \$220 million (existing assets 150, cash 50, tax shield 20).

Step 3 — (c) Price and shares repurchased. Equity is assets minus debt, spread over the 20 million shares still outstanding:

$$E_1 = A_1 - D = (220) - (50) = 170 \text{ million},$$
$$P_1 = \frac{E_1}{N_0} = \frac{170}{20} = 8.50, \tag{12.2}$$
$$n_R = \frac{D}{P_1} = \frac{(50)}{(8.50)} = 5.882353 \text{ million shares}. \tag{12.3}$$

Answer (c). $P_1 = \$8.50$; Kurz repurchases 5.88 million shares.

Step 4 — (d) After the repurchase. The cash is gone and equity falls by the \$50 million paid out:

Assets	(\$m)	Liabilities and equity	(\$m)
Existing assets	150	Debt	50
Tax shield	20	Equity	120
Total	170	Total	170

With $20 - 5.882353 = 14.117647$ million shares left, the price is

$$P_2 = \frac{120}{14.117647} = 8.50,$$

unchanged by the repurchase itself.

Answer (d). Assets 150 + 20 = 170 = debt 50 + equity 120 (\$m); share price \$8.50.

Intuition. The whole tax benefit, \$20 million or \$1 per share, is capitalised into the price at the announcement. Shareholders who sell in the buyback and those who stay share it equally, since the repurchase is at the fair price \$8.50.

Common mistakes.

- Repurchasing at \$7.50 (50/7.50 = 6.6667 million shares): once the plan is known the price already includes the tax shield.
- Leaving the \$50 million cash out of (b) (answering 170).
- Forgetting that the buyback itself does not change the price: only the announcement does.

Book answer. None in the book.

Ref: Berk Ch. 16, §16.3 "Debt and Taxes", pp. 473–479 (recapitalising to capture the tax shield).

13. Derive the MM adjusted cost of capital — Ross 16.19, p. 578

What is asked. Prove that, with corporate taxes only, Ross's after-tax WACC (CMPC) equals $R_U[1 - T_C(D/V)]$.

Approach. Combine three facts from Ross §16.4: the unlevered firm is a perpetuity $EBIT(1 - T_C)$ priced at R_U; *MM I with taxes*, $V_L = V_U + T_C D$; and the definition of the CMPC. Multiplying the CMPC by V turns it into the total after-tax cash flow paid to investors net of the tax saving, which is the unlevered cash flow. A second route goes through *MM II with taxes*.

Notation.

- $EBIT$: constant perpetual operating income, fully paid out; $T_C \in [0, 1)$: corporate tax rate (no personal taxes).
- D: perpetual riskless debt with rate R_D; E: levered equity value; $V = E + D = V_L$: levered firm value; V_U: unlevered value.

- R_U: unlevered cost of capital; R_E: cost of levered equity; CMPC $= \frac{E}{V}R_E + \frac{D}{V}R_D(1-T_C)$ (Ross's WACC). Assume $E > 0$, $V > 0$.

Claim. CMPC $= R_U\big[1 - T_C(D/V)\big]$.

Step 1 — Valuation identities (Ross §16.4). The unlevered firm's perpetual cash flow is $EBIT(1-T_C)$, and $V_U = EBIT(1-T_C)/R_U$; multiplying by R_U,

$$EBIT(1-T_C) = R_U V_U. \tag{13.1}$$

The shield $T_C R_D D$ is a perpetuity as risky as the debt, worth $T_C R_D D/R_D = T_C D$ (cf. (3.2)), so MM I with taxes is

$$V = V_U + T_C D. \tag{13.2}$$

Step 2 — Returns of the claims. Shareholders receive the after-tax income left after interest, $(EBIT - R_D D)(1-T_C)$, forever; debtholders receive $R_D D$, a perpetuity worth D at rate R_D. Valuing the equity stream as a perpetuity, $E = (EBIT - R_D D)(1-T_C)/R_E$, so

$$E\,R_E = (EBIT - R_D D)(1-T_C). \tag{13.3}$$

Step 3 — Multiply the CMPC by V. From the definition,

$$\begin{aligned}
V \cdot \text{CMPC} &= E\,R_E + D\,R_D(1-T_C) \\
&= \big[(EBIT - R_D D)(1-T_C)\big] + R_D D(1-T_C) && \text{(substituting (13.3))} \\
&= EBIT(1-T_C) - R_D D(1-T_C) + R_D D(1-T_C) && \text{(expanding)} \\
&= EBIT(1-T_C) \\
&= \big[R_U V_U\big] && \text{(substituting (13.1))} \\
&= R_U\big[V - T_C D\big] && \text{(substituting (13.2)).}
\end{aligned} \tag{13.4}$$

Step 4 — Divide by $V > 0$.

$$\text{CMPC} = R_U\,\frac{V - T_C D}{V} = R_U\Big[1 - T_C\frac{D}{V}\Big]. \qquad \blacksquare \tag{13.5}$$

Answer (proof). CMPC $= R_U\big[1 - T_C(D/V)\big]$, proved in (13.4)–(13.5).

Step 5 — Same result through MM II. First derive MM II with taxes. From (13.1) and (13.2), $EBIT(1-T_C) = R_U V_U = R_U(V - T_C D) = R_U(E + D - T_C D)$. Expand (13.3) and substitute:

$$\begin{aligned}
E\,R_E &= EBIT(1-T_C) - R_D D(1-T_C) \\
&= R_U(E + D - T_C D) - R_D D(1-T_C) && \text{(substituting)} \\
&= R_U E + R_U D(1-T_C) - R_D D(1-T_C) \\
&= R_U E + (R_U - R_D)D(1-T_C).
\end{aligned}$$

Dividing by E gives Ross's [16.4], $R_E = R_U + (R_U - R_D)\frac{D}{E}(1-T_C)$. Substitute it into the CMPC:

$$\begin{aligned}
\text{CMPC} &= \tfrac{E}{V}\big[R_U + (R_U - R_D)\tfrac{D}{E}(1-T_C)\big] + \tfrac{D}{V}R_D(1-T_C) \\
&= \tfrac{E}{V}R_U + \tfrac{D}{V}(R_U - R_D)(1-T_C) + \tfrac{D}{V}R_D(1-T_C) \\
&= \tfrac{E}{V}R_U + \tfrac{D}{V}R_U(1-T_C) \\
&= R_U\Big[\tfrac{E+D}{V} - T_C\tfrac{D}{V}\Big] = R_U\Big[1 - T_C\tfrac{D}{V}\Big],
\end{aligned}$$

since $E + D = V$.

Step 6 — Numerical check with Exercise 3. There $T_C D/V = 21{,}000/419{,}666.67 = 0.05003971$, so CMPC $= (0.15)(1 - 0.05003971) = 0.14249404$, and $(0.14249404)(419{,}666.67) = 59{,}800.00 =$

$EBIT(1-T_C)$, as (13.4) requires. Directly: equity income is $(92{,}000-5{,}400)(0.65)=56{,}290$, so $R_E=56{,}290/359{,}666.67=0.156506$ (MM II gives the same: $0.15+(0.06)(0.166821)(0.65)=0.15+0.006506$), and with $E/V=0.857029$, $D/V=0.142971$,

$$\text{CMPC}=(0.857029)(0.156506)+(0.142971)(0.09)(0.65)=0.134130+0.008364=0.142494.$$

Intuition. The levered firm's investors, taken together, receive the unlevered cash flow $EBIT(1-T_C)$ *plus* the tax saving; the CMPC nets the saving out of the debt cost, so it must discount the unlevered cash flow to the levered value. Since V exceeds V_U by T_CD, the rate falls below R_U by exactly R_UT_CD/V. The formula relies on fixed perpetual debt; under a constant leverage ratio the shield is riskier and the formula changes.

Common mistakes.

- Using MM II without taxes, $R_E=R_U+(R_U-R_D)D/E$: then the CMPC does not reduce to the claim.
- Confusing V with V_U in D/V: the weights are levered market values.
- Applying the formula to a firm that keeps D/V constant: it is derived for fixed perpetual debt.

Book answer. None in the local copy (selected answers are in the web-only Apêndice C).

Ref: Ross Cap. 16, §16.4, pp. 551–556 (Proposition I with taxes; taxes, WACC and Proposition II).

14. Miller's gain from leverage with personal taxes — Amaro de Matos 2.2.7 #1, p. 58

What is asked. Prove that, with corporate tax τ_c and personal taxes τ_s on equity income and τ_b on interest, the value that leverage creates for stockholders is $G_L=\left[1-(1-\tau_c)(1-\tau_s)/(1-\tau_b)\right]B$.

Approach. Use the *homemade-leverage* replication of MM, now counting every tax on each income stream (Miller 1977). An investor can copy the after-tax income of the levered firm's equity by holding the unlevered firm's equity and borrowing personally; since interest she pays is deductible at τ_b, she must borrow less than the firm's share of debt. No arbitrage then pins down S_l, and $G_L=V_l-V_u$ follows.

Notation.

- X: random operating income per period of two otherwise identical firms, u (unlevered) and l (levered), forever.
- B: market value of l's perpetual riskless debt, which pays rB per period; $r>0$ is the bond rate.
- $\tau_c,\tau_s,\tau_b\in[0,1)$: corporate tax rate (interest deductible), personal tax rate on equity income, personal tax rate on interest income.
- $S_u=V_u$, S_l: equity values; $V_l=S_l+B$; $\alpha\in(0,1]$: fraction held; L: personal loan; $G_L=V_l-V_u$: gain from leverage.

Assumption. As in Amaro's §2.2 and Appendix answer, individuals borrow and lend at the bond rate r, and personal interest is taxed (lending) or deductible (borrowing) at τ_b; there are no transaction costs and no arbitrage.

Step 1 — Payoff of the levered equity. The firm pays corporate tax on $X-rB$, and the holder pays τ_s on what she receives:

$$Y_l=\alpha(X-rB)(1-\tau_c)(1-\tau_s). \tag{14.1}$$

Step 2 — Payoff of the unlevered equity. The same taxes apply to the whole X:

$$Y_u=\alpha X(1-\tau_c)(1-\tau_s). \tag{14.2}$$

Step 3 — Size of the personal loan. A loan L at r costs rL in interest, of which $\tau_b rL$ comes back as a tax deduction, so $rL(1-\tau_b)$ per period after tax. Choose L so this equals the debt-service part of (14.1), $\alpha rB(1-\tau_c)(1-\tau_s)$:

$$\begin{aligned} rL(1-\tau_b) &= \alpha rB(1-\tau_c)(1-\tau_s) \\ L &= \alpha B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} \qquad \text{(dividing by } r(1-\tau_b)>0\text{).} \end{aligned} \tag{14.3}$$

Step 4 — Replication. Buying α of u and borrowing L pays, by (14.2) and (14.3),

$$\begin{aligned} Y_u - rL(1-\tau_b) &= \alpha X(1-\tau_c)(1-\tau_s) - \big[\alpha rB(1-\tau_c)(1-\tau_s)\big] \\ &= \alpha(X-rB)(1-\tau_c)(1-\tau_s) = Y_l, \end{aligned}$$

in every state and every period. Its cost today is $\alpha S_u - L$.

Step 5 — Law of one price. If $\alpha S_l > \alpha S_u - L$, a holder of l's shares sells them, builds the replica and keeps the difference with unchanged future income. If $\alpha S_l < \alpha S_u - L$, a holder of u's shares sells them, buys α of l's shares and lends L (after-tax interest $rL(1-\tau_b)$, restoring Y_u by Step 4 read backwards), again pocketing the difference. No arbitrage therefore requires

$$\begin{aligned} \alpha S_l &= \alpha S_u - L \\ \alpha S_l &= \alpha S_u - \alpha B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} && \text{(substituting (14.3))} \\ S_l &= S_u - B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b} && \text{(dividing by } \alpha\text{).} \end{aligned} \tag{14.4}$$

Step 6 — Firm values and the gain. Add B to both sides of (14.4) and use $V_l = S_l + B$, $V_u = S_u$:

$$\begin{aligned} V_l &= V_u + B - B\,\frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}, \\ G_L = V_l - V_u &= \Big[1 - \frac{(1-\tau_c)(1-\tau_s)}{1-\tau_b}\Big]B. \qquad \blacksquare \end{aligned} \tag{14.5}$$

The gain goes to the stockholders who decide to lever: they collect B at issue and keep shares worth S_l, a total of V_l instead of V_u.

Answer (proof). $G_L = \big[1 - (1-\tau_c)(1-\tau_s)/(1-\tau_b)\big]B$, proved in (14.5).

Step 7 — Special cases. (i) $\tau_s = \tau_b$ (including no personal taxes): $G_L = \tau_c B$, the MM (1963) result, Amaro (2.7). (ii) Miller's equilibrium (2.11), $(1-\tau_c)(1-\tau_s) = 1-\tau_b$: $G_L = 0$, so no single firm gains from leverage. (iii) $(1-\tau_c)(1-\tau_s) > 1-\tau_b$: $G_L < 0$, and the firm should not borrow. Illustration: with $\tau_c = 0.34$, $\tau_s = 0.15$, $\tau_b = 0.30$, $(0.66)(0.85)/0.70 = 0.561/0.70 = 0.801429$, so $G_L = 0.198571\,B$, well below $\tau_c B = 0.34\,B$.

Intuition. Debt moves a dollar of income from the corporate-plus-equity tax channel, where it keeps $(1-\tau_c)(1-\tau_s)$, to the interest channel, where it keeps $(1-\tau_b)$. Leverage creates value only to the extent that the second channel is less taxed than the first.

Common mistakes.

- Borrowing αB (the firm's share of debt) instead of L in (14.3): the personal deduction at τ_b changes the size of the loan that copies the firm's debt service.
- The sign of the loan: borrowing *reduces* the cost of the replica, so $\alpha S_l = \alpha S_u - L$.
- Concluding that $G_L = \tau_c B$ in general: that holds only when $\tau_s = \tau_b$.

Book answer. The final expression matches the book's answer (p. 255). The intermediate line printed there, "$\alpha S_l = \alpha S_u + \alpha B\frac{(1-\tau_c)(1-\tau_s)}{(1-\tau_b)}$", has a sign typo: it must be a minus, as in (14.4), since the investor *borrows* that amount and the book's own next line, $V_l = V_u + B[1 - \cdots]$, follows only with the minus.

Ref: Amaro de Matos Ch. 2, §2.2 "Personal and Corporate Taxation", pp. 48–52 (Miller equilibrium, G_L on p. 52); Appendix C.2.2, p. 255.

15. MM I with perpetual debt, and when borrowing and lending rates differ — Amaro de Matos 2.1.5 #3–#4, p. 48

What is asked. #3: Prove MM I in the original 1958 setting (perpetual debt paying rD each period, income measured as a perpetual average X), both without taxes ($V_l = V_u$) and with deductible interest ($V_l = V_u + \tau B$). #4: What becomes of MM I if individuals borrow at $\bar{r}$ and lend at $\underline{r}$, with $\bar{r} > r > \underline{r}$, while firms borrow at r.

Approach. Both parts use the *homemade-leverage* arbitrage. For each firm's equity, build a portfolio from the other firm's securities plus personal borrowing or lending that pays the same income in every state; no arbitrage then forces the two to cost the same, which gives two inequalities whose combination is MM I. In #4 personal borrowing is dearer and lending cheaper than corporate debt, so the replicas cost more and the two inequalities no longer meet: they leave a band.

Notation.

- X: random operating income per period (the time average of the stream), identical for firms u (unlevered) and l (levered); any distribution.
- B: market value of l's perpetual riskless debt, paying rB per period; r: corporate interest rate.
- $\tau \in [0, 1)$: corporate tax rate, interest deductible ($\tau = 0$ is the no-tax case); no personal taxes.
- $V_u = S_u$, $V_l = S_l + B$; $\alpha \in (0, 1]$: fraction of a firm held by an investor.
- #4 only: $\bar{r}$ and $\underline{r}$: individual borrowing and lending rates, $\bar{r} > r > \underline{r} > 0$.

Assumption. In #3 individuals borrow and lend at r, as firms do; there are no transaction costs. In #4, as in the book's answer, $\tau = 0$ and individuals can lend only at $\underline{r}$ (they cannot buy l's bonds at yield r).

#3 — MM I with and without taxes

Step 1 — Payoffs. Shareholders of u pay tax on all of X; those of l pay tax on X net of interest. Each period they receive

$$Y_u = X(1-\tau), \qquad Y_l = (X - rB)(1-\tau). \tag{15.1}$$

Step 2 — Replicating l's equity. Buy α of u and borrow $\alpha(1-\tau)B$ at r. By (15.1) the income is

$$\alpha X(1-\tau) - r\,\alpha(1-\tau)B = \alpha(X - rB)(1-\tau) = \alpha Y_l,$$

at a cost of $\alpha S_u - \alpha(1-\tau)B$ today.

Step 3 — Upper bound. If $S_l > S_u - (1-\tau)B$, a holder of α of l's shares sells them for αS_l, builds the replica of Step 2 and keeps, with identical future income,

$$\begin{aligned} \alpha\big[S_l - S_u + (1-\tau)B\big] &= \alpha\big[(V_l - B) - V_u + (1-\tau)B\big] \qquad (\text{using } S_l = V_l - B,\ S_u = V_u) \\ &= \alpha\big[V_l - V_u - B + B - \tau B\big] \\ &= \alpha\big[V_l - V_u - \tau B\big] > 0, \end{aligned}$$

an arbitrage. Hence $V_l \le V_u + \tau B$.

Step 4 — Replicating u's equity. Buy α of l's shares and $\alpha(1-\tau)$ of its debt (or lend $\alpha(1-\tau)B$ at r):

$$\alpha(X - rB)(1-\tau) + \alpha(1-\tau)rB = \alpha X(1-\tau) = \alpha Y_u,$$

at a cost of $\alpha\big[S_l + (1-\tau)B\big]$.

Step 5 — Lower bound. If $S_u > S_l + (1-\tau)B$, then substituting $S_u = V_u$, $S_l = V_l - B$ gives $V_u > V_l - B + B - \tau B$, i.e. $V_l < V_u + \tau B$. A holder of u's shares then sells them, buys the replica of Step 4 and keeps $\alpha[V_u + \tau B - V_l] > 0$. Hence $V_l \ge V_u + \tau B$.

Step 6 — MM I. Steps 3 and 5 give

$$V_l = V_u + \tau B \quad \text{(with taxes)}, \qquad V_l = V_u \quad (\tau = 0). \qquad \blacksquare \tag{15.2}$$

Answer (#3). $V_l = V_u + \tau B$ with deductible interest; $V_l = V_u$ without taxes.

Book answer. Matches the book's answer (pp. 252–253), $V_u + \tau B = V_l$. The book's switching portfolio invests all the proceeds in the other firm instead of replicating exactly, which leads to the same two inequalities.

#4 — Different borrowing and lending rates ($\tau = 0$)

Step 7 — Replicating l's equity with costly borrowing. The replica must still hold α of u (the only other claim on X), so its debt service has to equal $\alpha r B$ at the rate $\bar r$:

$$\bar r\,\bar L = \alpha r B \quad \Longrightarrow \quad \bar L = \alpha B\,\frac{r}{\bar r}, \tag{15.3}$$

which gives income $\alpha X - \bar r \bar L = \alpha X - \alpha r B = \alpha Y_l$ at a cost of $\alpha S_u - \bar L = \alpha\big(S_u - B\,r/\bar r\big)$. (Borrowing the full αB would cost $\alpha \bar r B > \alpha r B$ per period: not a replica.)

Step 8 — Upper bound. No arbitrage against an overvalued l requires $\alpha S_l \le \alpha\big(S_u - B\,r/\bar r\big)$, i.e. $S_l \le S_u - B\,r/\bar r$. Add B to both sides and use $V_l = S_l + B$, $V_u = S_u$:

$$V_l \le V_u + B - B\,\frac{r}{\bar r} = V_u + B\Big(1 - \frac{r}{\bar r}\Big) = V_u + B\,\frac{\bar r - r}{\bar r}. \tag{15.4}$$

Step 9 — Replicating u's equity with poor lending. Buy α of l's shares and lend $\underline{L}$ at $\underline{r}$ with

$$\underline{r}\,\underline{L} = \alpha r B \quad \Longrightarrow \quad \underline{L} = \alpha B\,\frac{r}{\underline{r}} > \alpha B, \tag{15.5}$$

giving income $\alpha(X - rB) + \alpha r B = \alpha Y_u$ at a cost of $\alpha\big(S_l + B\,r/\underline{r}\big)$.

Step 10 — Lower bound. No arbitrage against an overvalued u requires $S_u \le S_l + B\,r/\underline{r}$. Substitute $S_u = V_u$ and $S_l = V_l - B$:

$$\begin{aligned} V_u &\le V_l - B + B\,\frac{r}{\underline{r}} \\ V_l &\ge V_u + B - B\,\frac{r}{\underline{r}} = V_u - B\Big(\frac{r}{\underline{r}} - 1\Big) = V_u - B\,\frac{r - \underline{r}}{\underline{r}}. \end{aligned} \tag{15.6}$$

Step 11 — What happens to MM1. Combining (15.4) and (15.6),

$$V_u - B\,\frac{r - \underline{r}}{\underline{r}} \;\le\; V_l \;\le\; V_u + B\,\frac{\bar r - r}{\bar r}. \qquad \blacksquare \tag{15.7}$$

MM1 no longer pins down V_l: arbitrage only confines it to a band around V_u whose width grows with the debt B and with the spreads. With $\bar r = r + \epsilon$, $\underline{r} = r - \epsilon$ the band is $V_u - \epsilon B/(r-\epsilon) \le V_l \le V_u + \epsilon B/(r+\epsilon)$, and it collapses to $V_l = V_u$ as $\epsilon \to 0$. Leverage can now raise firm value (up to the

upper bound): the firm borrows more cheaply than individuals, so corporate leverage is a service investors cannot fully copy. If individuals could buy l's bonds at yield r, Step 4 would apply and the lower bound would tighten to $V_l \geq V_u$.

Answer (#4). MM1 fails as an equality: V_l is only confined to the band (15.7) around V_u, which shrinks to $V_l = V_u$ as the spreads vanish.

Intuition. MM1 rests on investors being able to copy corporate leverage on the same terms. When personal borrowing is dearer and lending cheaper than corporate debt, homemade leverage is an imperfect substitute: arbitrage still limits mispricing, but only to within the cost of the personal-borrowing wedge.

Common mistakes.

- In #3 with taxes, borrowing αB instead of $\alpha(1-\tau)B$: the firm's interest is tax-deductible, so its equity bears only $(1-\tau)rB$ of debt service.
- In #4, borrowing the full αB at $\bar{r}$ and calling it a replica: the debt service then differs, so the incomes do not match state by state.
- Concluding that MM1 "fails completely": arbitrage still bounds V_l, only less tightly.

Book answer. Differs from the book's printed band (p. 254), $V_u(1-\epsilon B_l/X) \leq V_l \leq V_u(1+\epsilon B_l/X)$. The qualitative conclusion is the same (a band around V_u, proportional to ϵB, vanishing at $\epsilon = 0$), but the bounds differ. The book's portfolios keep the investor's capital fixed (all proceeds go into the other firm, and the full principal B_l is borrowed or lent), so the two incomes differ by a multiple of the random X minus a fixed amount. Its inequality therefore holds only for a given value of X (or in expectation), not state by state. The portfolios above match incomes exactly, so (15.7) is a true no-arbitrage band for any distribution of X. The two coincide only if X/V_u happens to equal $\bar{r}$ (upper bound) or $\underline{r}$ (lower bound).

Ref: Amaro de Matos Ch. 2, §2.1.1–2.1.3, pp. 40–46 (MM1, MM2, MM with taxes); Appendix C.2.1, pp. 252–254.

16. Bank capital without taxes: does more equity lower ROE and raise funding costs? — Original exercise

Original exercise (list, Exercise 16), built on Topic 2B slides 46–47 and Topic 2A slides 89 and 97–102. Amounts in $ billion.

What is asked. (a) The balance sheet at the two equity ratios and a check that deposits are really safe. (b) What shareholders earn per dollar invested (ROE) in each state, its mean and its risk, and the reward per unit of risk. (c) The return shareholders *require* at each ratio (MM II), whether it matches what they expect to earn, and whether the overall cost of funding (WACC) or the bank's value moves. (d) How a shareholder can rebuild the high-leverage ROE herself by borrowing. (e)–(f) Whether Statements 1 and 2 of slide 47 survive these numbers.

Approach. The bank is a firm in a perfect market, so MM applies unchanged: the loans' cash flows and their required return r_A do not depend on how they are funded (*MM I*). Leverage only redistributes the same risk: with less equity, each dollar of equity carries more of the asset risk and therefore must earn more (*MM II*). ROE compares returns on claims of different risk, so a lower ROE at higher capital is not a loss. *Homemade leverage* shows the old ROE remains available to any investor who wants it.

Notation.

- $A = 100$: market value of the loans at $t = 0$; $\tilde{r}_A \in \{9\%, 3\%\}$: return on assets (good, bad state), probability $\frac{1}{2}$ each; $r_A = 6\%$: required return on assets; σ_A: standard deviation of $\tilde{r}_A$.
- $e = E/A$: equity ratio; E, $D = A - E$: equity and deposits; $r_D = r_f = 4\%$: deposit (risk-free) rate.

- Π_s: payoff to shareholders at $t=1$ in state $s \in \{g, b\}$; $ROE_s = \Pi_s/E - 1$; r_E: required return on equity; r_{WACC}: weighted average cost of capital.

Step 1 — (a) Balance sheets and the repayment owed. The equity ratio fixes $E = eA$, deposits are the rest, and depositors are owed principal plus interest:

$$\begin{aligned} e = 4\%: &\quad E = (0.04)(100) = 4, \quad D = 100 - 4 = 96, \quad D(1+r_f) = (96)(1.04) = 99.84, \\ e = 20\%: &\quad E = (0.20)(100) = 20, \quad D = 100 - 20 = 80, \quad D(1+r_f) = (80)(1.04) = 83.2. \end{aligned} \tag{16.1}$$

Step 2 — (a) Deposits are safe. The worst case for depositors is the bad state, where the loans pay

$$A(1+0.03) = (100)(1.03) = 103. \tag{16.2}$$

Since $103 > 99.84$ and $103 > 83.2$, depositors are paid in full in both states at both ratios, so the deposits are risk-free and $r_D = r_f$ is the right rate. In the good state the loans pay $(100)(1.09) = 109$.

Answer (a). $e = 4\%$: $E = 4$, $D = 96$, owed 99.84; $e = 20\%$: $E = 20$, $D = 80$, owed 83.2; both below the bad-state asset payoff of 103, so deposits are risk-free.

Step 3 — (b) Shareholders' payoffs. Shareholders receive what is left after depositors:

$$\begin{aligned} e = 4\%: &\quad \Pi_g = 109 - 99.84 = 9.16, \qquad \Pi_b = 103 - 99.84 = 3.16, \\ e = 20\%: &\quad \Pi_g = 109 - 83.2 = 25.8, \qquad \Pi_b = 103 - 83.2 = 19.8. \end{aligned}$$

Step 4 — (b) ROE in each state. Divide each payoff by the equity invested and subtract 1:

$$\begin{aligned} e = 4\%: &\quad ROE_g = \frac{9.16}{4} - 1 = 2.29 - 1 = 129\%, \qquad ROE_b = \frac{3.16}{4} - 1 = 0.79 - 1 = -21\%, \\ e = 20\%: &\quad ROE_g = \frac{25.8}{20} - 1 = 1.29 - 1 = 29\%, \qquad ROE_b = \frac{19.8}{20} - 1 = 0.99 - 1 = -1\%. \end{aligned}$$

The same numbers follow from $ROE_s = \tilde{r}_{A,s} + \frac{D}{E}(\tilde{r}_{A,s} - r_f)$: at $e = 4\%$, $D/E = 96/4 = 24$ and $ROE_g = 9\% + 24(9\% - 4\%) = 9\% + 120\% = 129\%$.

Step 5 — (b) Mean, risk and reward per unit of risk. With two equally likely states the mean is the average and the standard deviation is half the gap between the two outcomes:

$$\begin{aligned} e = 4\%: &\quad \mathbb{E}[ROE] = \tfrac{1}{2}(129\%) + \tfrac{1}{2}(-21\%) = 54\%, \\ &\quad \sigma(ROE) = \tfrac{1}{2}\,|129\% - (-21\%)| = \tfrac{1}{2}(150\%) = 75\%, \\ e = 20\%: &\quad \mathbb{E}[ROE] = \tfrac{1}{2}(29\%) + \tfrac{1}{2}(-1\%) = 14\%, \\ &\quad \sigma(ROE) = \tfrac{1}{2}\,|29\% - (-1\%)| = \tfrac{1}{2}(30\%) = 15\%. \end{aligned} \tag{16.3}$$

For comparison, $\sigma_A = \frac{1}{2}|9\% - 3\%| = 3\%$, and $\sigma(ROE) = (A/E)\,\sigma_A$: $(25)(3\%) = 75\%$ and $(5)(3\%) = 15\%$. The reward per unit of risk is

$$e = 4\%: \quad \frac{54\% - 4\%}{75\%} = \frac{50}{75} = 0.6667, \qquad e = 20\%: \quad \frac{14\% - 4\%}{15\%} = \frac{10}{15} = 0.6667. \tag{16.4}$$

Answer (b). $e = 4\%$: ROE 129% / −21%, $\mathbb{E}[ROE] = 54\%$, $\sigma = 75\%$; $e = 20\%$: ROE 29% / −1%, $\mathbb{E}[ROE] = 14\%$, $\sigma = 15\%$. The ratio is 0.6667 at both: expected ROE falls exactly in proportion to risk.

Step 6 — (c) Required return on equity (MM II). With safe debt, $r_E = r_A + \frac{D}{E}(r_A - r_D)$:

$$\begin{aligned} e = 4\%: &\quad r_E = 6\% + \Big(\frac{96}{4}\Big)(6\% - 4\%) = 6\% + (24)(2\%) = 6\% + 48\% = 54\%, \\ e = 20\%: &\quad r_E = 6\% + \Big(\frac{80}{20}\Big)(6\% - 4\%) = 6\% + (4)(2\%) = 6\% + 8\% = 14\%. \end{aligned} \tag{16.5}$$

These equal the expected ROEs in (16.3): at both ratios shareholders expect to earn exactly what their risk requires, so their claim has zero NPV either way. Indeed the equity is worth $\mathbb{E}[\Pi]/(1+r_E)$: $\frac{1}{2}(9.16+3.16)/1.54 = 6.16/1.54 = 4$ and $\frac{1}{2}(25.8+19.8)/1.14 = 22.8/1.14 = 20$, the amounts invested.

Step 7 — (c) WACC and value. Weight each required return by its share of funding:

$$\begin{aligned} e = 4\%: &\quad r_{WACC} = (0.04)(54\%) + (0.96)(4\%) = 2.16\% + 3.84\% = 6\%, \\ e = 20\%: &\quad r_{WACC} = (0.20)(14\%) + (0.80)(4\%) = 2.80\% + 3.20\% = 6\%. \end{aligned} \tag{16.6}$$

Both equal r_A. The bank's value is $E + D = 4 + 96 = 20 + 80 = 100 = A$ at both ratios.

Answer (c). $r_E = 54\%$ at 4% equity and 14% at 20% equity, equal to the expected ROEs; $r_{WACC} = 6\%$ at both; the bank's value stays 100.

Step 8 — (d) Homemade leverage. One dollar of the 4% bank's equity carries $A/E = 1/0.04 = 25$ dollars of loans; one dollar of the 20% bank's equity carries $1/0.20 = 5$. To rebuild 25 dollars of loan exposure per dollar of her own money, she buys $25/5 = 5$ dollars of the 20% bank's equity, financed with her \$1 plus $5 - 1 = 4$ dollars borrowed at r_f. Her return in each state:

$$\begin{aligned} \text{good:} &\quad \frac{(5)(1.29) - (4)(1.04)}{1} - 1 = 6.45 - 4.16 - 1 = 1.29 = 129\%, \\ \text{bad:} &\quad \frac{(5)(0.99) - (4)(1.04)}{1} - 1 = 4.95 - 4.16 - 1 = -0.21 = -21\%. \end{aligned} \tag{16.7}$$

Answer (d). Buy \$5 of the 20% bank's equity per \$1 of her own money, borrowing \$4 at 4%; she gets 129% and −21%, exactly the 4% bank's ROE.

Step 9 — (e) Statement 1. Expected ROE does fall, from 54% to 14%, but risk falls in the same proportion (16.4), the required return falls by the same amount (16.5), the value of the shareholders' claim equals what they paid at both ratios, and the high-ROE payoff is still available through (16.7).

Answer (e). False. A lower ROE is the price of lower risk, not a loss to shareholders.

Step 10 — (f) Statement 2. The executive's computation keeps r_E at its 4%-equity level:

$$(0.20)(54\%) + (0.80)(4\%) = 10.8\% + 3.2\% = 14\%. \tag{16.8}$$

But r_E is not a constant: by MM II it falls to 14% when equity rises to 20%, and the correct WACC (16.6) stays at 6%. Equity has a higher required return than debt at every ratio, yet replacing debt with equity does not raise the funding cost, because it makes all equity less risky.

Answer (f). The executive's number is 14%; the error is holding r_E fixed at 54%. Statement 2 is false: the funding cost stays at 6%.

Intuition. The loans' risk is fixed by the loans, not by the liability side. Leverage concentrates that risk on a thin layer of equity, which then demands a high return; more equity spreads the same risk over more dollars, so each dollar demands less. ROE is a return on a risky claim, and comparing ROEs across leverage levels compares claims of different risk.

Common mistakes.

- Judging shareholder welfare by expected ROE instead of by the value of their claim (or ROE per unit of risk).
- Using a fixed r_E when leverage changes, as in (16.8).
- Forgetting to check that deposits are safe; MM II in the linear form needs $r_D = r_f$.

Ref: Prof. Ricca, Topic 2B slides 46–47 (bank capital debate, Statements 1–2); Topic 2A slide 89 (homemade leverage) and slides 97–102 (MM II, r_{WACC} independent of D and E, conservation of value).

17. Bank capital with corporate taxes: the lost interest tax shield — Original exercise

Original exercise (list, Exercise 17), built on Topic 2B slides 7–18 and 47. Amounts in $ billion, flows per year.

What is asked. (a) The bank's market-value balance sheet at 5% and 20% equity when interest is tax-deductible. (b) What the move to 20% requires (debt retired) and what it costs the bank's current owners. (c) How the after-tax WACC and the required return on equity change, with two consistency checks. (d) What happens to the government's tax claim and to the total value of all claims. (e) Whether the lost tax shield (Statement 3) is a private cost, a social cost, or both.

Approach. With constant perpetual safe debt, *MM I with taxes* gives $V_L = V_U + \tau D$ (slides 7–8): the interest tax shield is the only thing leverage adds. Because the requirement is stated as a share of V_L, which itself depends on D, part (a) is a fixed point solved in one line. Part (d) counts the government as a third claimant: the pie X is split among shareholders, debtholders and the government, and leverage only moves value between them.

Notation.

- $X = 7.5$: expected pre-tax cash flow per year (perpetual); $\tau = 20\%$: tax rate; $r_U = 6\%$: unlevered cost of capital; $V_U = (1-\tau)X/r_U = 100$.
- D: perpetual constant safe debt, $r_D = 4\%$; E: equity; $V_L = D + E$; $e = E/V_L$; $V_{ITS} = \tau D$.
- r_E: required return on equity; r_{WACC}: after-tax WACC; G: present value of the government's tax claim.

Step 1 — (a) Solve for V_L. Substitute $D = (1-e)V_L$ into $V_L = V_U + \tau D$ and collect V_L:

$$\begin{aligned} V_L &= V_U + \tau(1-e)V_L \\ V_L\big[1-\tau(1-e)\big] &= V_U \\ V_L &= \frac{V_U}{1-\tau(1-e)}. \end{aligned} \tag{17.1}$$

Step 2 — (a) Balance sheet at $e = 5\%$.

$$\begin{aligned} V_L &= \frac{100}{1-(0.20)(0.95)} = \frac{100}{1-0.19} = \frac{100}{0.81} = 123.4568, \\ D &= (0.95)(123.4568) = 117.2840, \qquad E = (0.05)(123.4568) = 6.1728, \\ V_{ITS} &= (0.20)(117.2840) = 23.4568 = V_L - V_U. \end{aligned}$$

Step 3 — (a) Balance sheet at $e = 20\%$.

$$\begin{aligned} V_L &= \frac{100}{1-(0.20)(0.80)} = \frac{100}{1-0.16} = \frac{100}{0.84} = 119.0476, \\ D &= (0.80)(119.0476) = 95.2381, \qquad E = (0.20)(119.0476) = 23.8095, \\ V_{ITS} &= (0.20)(95.2381) = 19.0476 = V_L - V_U. \end{aligned}$$

Answer (a). $e = 5\%$: $V_L = 123.457$, $D = 117.284$, $E = 6.173$, $V_{ITS} = 23.457$; $e = 20\%$: $V_L = 119.048$, $D = 95.238$, $E = 23.810$, $V_{ITS} = 19.048$.

Step 4 — (b) Debt retired and the annual tax saving. The debt retired equals the new equity raised:

$$\Delta D = 117.2840 - 95.2381 = 22.0459.$$

The annual interest tax saving $\tau r_D D$ is

$$e = 5\%: \ (0.20)(0.04)(117.2840) = 0.9383, \qquad e = 20\%: \ (0.20)(0.04)(95.2381) = 0.7619,$$

a fall of $0.9383 - 0.7619 = 0.1764$ per year.

Step 5 — (b) Loss to existing shareholders. New shareholders pay a fair price, so they pay exactly what their shares are worth and neither gain nor lose. The fall in V_L therefore lands entirely on the existing shareholders:

$$\Delta V_L = 123.4568 - 119.0476 = 4.4092 = \tau\,\Delta D = (0.20)(22.0459), \tag{17.2}$$

which is also the perpetual annual loss discounted at r_D: $0.1764/0.04 = 4.41$. The existing shareholders keep $23.8095 - 22.0459 = 1.7637 = 6.1728 - 4.4092$, so the loss as a fraction of their equity is

$$\frac{4.4092}{6.1728} = 0.7143 = 71.43\%.$$

Answer (b). Retire 22.046 of debt; the annual tax saving falls from 0.938 to 0.762; existing shareholders lose 4.409, i.e. 71.43% of their pre-change equity.

Step 6 — (c) After-tax WACC. Constant-debt formula $r_{WACC} = (1-\tau\frac{D}{V_L})r_U$, with $D/V_L = 1-e$:

$$\begin{aligned}
e = 5\%: \quad & r_{WACC} = \big(1 - (0.20)(0.95)\big)(6\%) = (0.81)(6\%) = 4.86\%, \\
& \frac{(1-\tau)X}{r_{WACC}} = \frac{6}{0.0486} = 123.4568, \\
e = 20\%: \quad & r_{WACC} = \big(1 - (0.20)(0.80)\big)(6\%) = (0.84)(6\%) = 5.04\%, \\
& \frac{(1-\tau)X}{r_{WACC}} = \frac{6}{0.0504} = 119.0476.
\end{aligned}$$

Both reproduce V_L from Steps 2–3. The after-tax WACC rises by $5.04\% - 4.86\% = 0.18$ percentage points.

Step 7 — (c) Required return on equity. The balance-sheet identity (slide 10) multiplied by V_L, with $r_{ITS} = r_D$ and $V_{ITS} = \tau D$, is $E\,r_E + D\,r_D = V_U r_U + \tau D\,r_D$. Substitute $V_U = V_L - \tau D = E + (1-\tau)D$ and solve for r_E:

$$\begin{aligned}
E\,r_E &= \big[E + (1-\tau)D\big]r_U + \tau D\,r_D - D\,r_D \\
&= E\,r_U + (1-\tau)D\,r_U - (1-\tau)D\,r_D \\
r_E &= r_U + (1-\tau)\frac{D}{E}(r_U - r_D).
\end{aligned} \tag{17.3}$$

With $D/E = 0.95/0.05 = 19$ and $D/E = 0.80/0.20 = 4$:

$$\begin{aligned}
e = 5\%: \quad & r_E = 6\% + (0.80)(19)(2\%) = 6\% + 30.4\% = 36.4\%, \\
e = 20\%: \quad & r_E = 6\% + (0.80)(4)(2\%) = 6\% + 6.4\% = 12.4\%.
\end{aligned}$$

Check with the after-tax WACC formula (slide 16):

$$\begin{aligned}
e = 5\%: \quad & (0.05)(36.4\%) + (0.95)(4\%)(0.80) = 1.82\% + 3.04\% = 4.86\%, \\
e = 20\%: \quad & (0.20)(12.4\%) + (0.80)(4\%)(0.80) = 2.48\% + 2.56\% = 5.04\%.
\end{aligned}$$

Answer (c). After-tax WACC 4.86% → 5.04% (it gives back V_L); r_E 36.4% → 12.4%; both checks hold.

Step 8 — (d) The government's claim. Discount τX at r_U and $\tau r_D D$ at r_D:

$$G = \frac{\tau X}{r_U} - \frac{\tau r_D D}{r_D} = \frac{(0.20)(7.5)}{0.06} - \tau D = 25 - \tau D, \tag{17.4}$$

$$\begin{aligned}
e = 5\%: \quad & G = 25 - 23.4568 = 1.5432, \qquad V_L + G = 123.4568 + 1.5432 = 125, \\
e = 20\%: \quad & G = 25 - 19.0476 = 5.9524, \qquad V_L + G = 119.0476 + 5.9524 = 125.
\end{aligned}$$

The total $125 = X/r_U = 7.5/0.06$ is the pre-tax value of the assets, independent of D. The government gains $5.9524 - 1.5432 = 4.4092$, exactly the shareholders' loss in (17.2).

Answer (d). $G = 1.543$ at 5% and 5.952 at 20%; $V_L + G = 125$ at both; the taxpayers gain the 4.409 the shareholders lose.

Step 9 — (e) Statement 3. For the bank's shareholders the statement is true: they lose the subsidy 4.409 and the after-tax WACC rises. For society it is a transfer, not a cost: the total value of the claims on the assets stays at 125, and the government's extra revenue can fund spending or lower taxes elsewhere. The lost tax shield shows that the tax code subsidises bank debt, not that equity uses up resources.

Answer (e). True privately, false socially: the lost tax shield is a transfer from bank shareholders to taxpayers, so it does not make equity socially expensive.

Intuition. Taxes add a third claimant to the bank. Debt lowers what the government takes, and that is the only reason it raises V_L. A capital requirement moves that slice back to the government. Whether banks should get the subsidy is a separate tax-policy question; it is no reason to keep leverage high.

Common mistakes.

- Treating e as E/V_U instead of E/V_L, or computing $D = (1-e)V_U$; the requirement is a fixed point in V_L (17.1).
- Counting the lost tax shield as a loss to society; it reappears in G.
- Holding r_E at 36.4% when equity rises (the Statement 2 mistake, again).

Ref: Prof. Ricca, Topic 2B slides 7–8 (MM I with taxes, $V_{ITS} = \tau D$), slides 9–11 (market-value balance sheet, $r_{ITS} = r_D$ for constant debt), slides 16–18 (after-tax WACC, constant-debt formula), slide 47 (Statement 3).

18. The failure externality and the socially optimal capital ratio — Original exercise

Original exercise (list, Exercise 18), built on Topic 2B slides 44–47. All amounts per \$1 of bank assets per year.

What is asked. (a) How likely failure is, and how much it costs society in expectation, at a thin and a thick capital ratio. (b) Society's cost of the funding mix, why only two terms belong in it, and the ratio that minimises it. (c) What the bank chooses on its own, and how its private cost of more capital compares with society's gain. (d) How the slide-47 fallacies (Statements 2 and 3) bias a planner's "optimal" ratio downward. (e) How e^* responds to the social loss of a failure, the value of liquidity and asset risk.

Approach. This is the *static trade-off* of slides 44–45 seen from society's side: the marginal social benefit of debt (liquidity services) is set equal to the marginal expected cost of failure. By Exercises 16–17 (MM without and with taxes), neither the equity premium nor the tax shield is a social cost, so they are left out. The failure cost is a *negative externality* (slide 47): the bank does not bear it, which is why the private and social optima differ.

Notation.

- $e \in [0,1]$: equity ratio; $1-e$: deposits per \$1 of assets; $\ell \geq 0$: annual loss rate, $\Pr(\ell > x) = \exp(-x/\theta)$, $\theta = 0.01$.
- $p(e) = \Pr(\ell > e)$: annual failure probability; $\kappa = 0.50$: social loss of a failure per \$1 of assets; $F(e) = \kappa p(e)$.
- $\lambda = 0.005$: liquidity value per \$1 of deposits per year; $s = 0.008$: tax saving per \$1 of deposits per year.
- $S(e)$: social cost of the funding mix; e^*: its minimiser; $\underline{e}$: regulatory minimum.

Step 1 — (a) Failure probability and expected failure cost. The bank fails when $\ell > e$, so $p(e) = \exp(-e/\theta)$:

$$\begin{aligned} e = 4\%: \quad & p = \exp(-0.04/0.01) = \exp(-4) = 0.018316, \\ & F = (0.50)(0.018316) = 0.0091578, \\ e = 20\%: \quad & p = \exp(-0.20/0.01) = \exp(-20) = 2.06 \times 10^{-9}, \\ & F = (0.50)(2.06 \times 10^{-9}) = 1.03 \times 10^{-9}. \end{aligned} \tag{18.1}$$

Answer (a). $p(4\%) = 1.83\%$ a year and $F(4\%) = 0.916\%$ of assets a year; at 20% failure is practically impossible ($p = 2.06 \times 10^{-9}$, $F = 1.03 \times 10^{-9}$).

Step 2 — (b) The social cost function. Deposits yield a social benefit $\lambda(1-e) = \lambda - \lambda e$ a year, so relative to $e = 0$ each unit of equity forgoes λ of liquidity services. Adding the expected failure cost and dropping the constant $-\lambda$:

$$S(e) = \lambda e + \kappa \exp(-e/\theta). \tag{18.2}$$

The equity premium is absent because, by MM (Exercise 16), it only compensates shareholders for the risk they bear and the total required return on the assets does not change. The tax saving s is absent because it is a transfer from the government (Exercise 17): what the bank gains, taxpayers lose.

Step 3 — (b) First- and second-order conditions. Differentiate (18.2), using $\frac{d}{de}\exp(-e/\theta) = -\frac{1}{\theta}\exp(-e/\theta)$:

$$\begin{aligned} S'(e) &= \lambda - \frac{\kappa}{\theta}\exp(-e/\theta) = 0, \\ S''(e) &= \frac{\kappa}{\theta^2}\exp(-e/\theta) > 0, \end{aligned} \tag{18.3}$$

so the stationary point is the global minimum. Solve (18.3):

$$\begin{aligned} \exp(-e^*/\theta) &= \frac{\lambda\theta}{\kappa} \\ -\frac{e^*}{\theta} &= \ln\frac{\lambda\theta}{\kappa} \\ e^* &= \theta \ln\frac{\kappa}{\lambda\theta}. \end{aligned} \tag{18.4}$$

$e^* > 0$ because $\kappa/(\lambda\theta) > 1$. Numerically,

$$\frac{\kappa}{\lambda\theta} = \frac{0.50}{(0.005)(0.01)} = \frac{0.50}{0.00005} = 10{,}000, \qquad e^* = (0.01)\ln(10{,}000) = (0.01)(9.2103) = 0.092103.$$

By (18.4) the failure probability at the optimum is $p(e^*) = \lambda\theta/\kappa = 0.00005/0.50 = 0.0001$, once in 10,000 years.

Step 4 — (b) Social cost at three ratios.

$$\begin{aligned} S(4\%) &= (0.005)(0.04) + 0.0091578 = 0.0002 + 0.0091578 = 0.0093578, \\ S(e^*) &= (0.005)(0.092103) + (0.50)(0.0001) = 0.00046052 + 0.00005 = 0.00051052, \\ S(20\%) &= (0.005)(0.20) + 1.03 \times 10^{-9} = 0.0010000. \end{aligned}$$

Answer (b). $e^* = \theta\ln(\kappa/(\lambda\theta)) = 9.21\%$, with failure probability 0.01% a year; $S(4\%) = 0.93578\%$, $S(e^*) = 0.05105\%$, $S(20\%) = 0.10000\%$ of assets a year.

Step 5 — (c) The bank's choice. The shareholders' private cost $P(e) = (\lambda + s)e = 0.013\,e$ has $P'(e) = 0.013 > 0$ at every e: each extra unit of equity only costs them, and the failure cost it would

avoid falls on others. The bank picks the lowest ratio allowed: $e = 0$ unregulated, $e = \underline{e} = 4\%$ under the minimum. Moving from 4% to e^*:

$$\text{private cost} = (0.013)(0.092103 - 0.04) = (0.013)(0.052103) = 0.00067734,$$
$$\text{social gain} = S(4\%) - S(e^*) = 0.0093578 - 0.00051052 = 0.0088473.$$

Of the private cost, $(0.005)(0.052103) = 0.00026052$ is the liquidity loss (already inside S) and $(0.008)(0.052103) = 0.00041683$ is tax paid to the government.

Answer (c). The bank chooses $e = 0$ unregulated and exactly 4% under the minimum. The move to e^* costs shareholders 0.0677% of assets a year and saves society 0.8847%, about 13 times more.

Step 6 — (d) The planner's mistakes. Each mistake replaces λ in (18.4):

$$\text{(i)}\quad \lambda + s = 0.013: \quad e = (0.01)\ln\frac{0.50}{(0.013)(0.01)} = (0.01)\ln(3{,}846.15) = (0.01)(8.2548) = 0.082548,$$
$$\text{(ii)}\quad 0.10: \quad e = (0.01)\ln\frac{0.50}{(0.10)(0.01)} = (0.01)\ln(500) = (0.01)(6.2146) = 0.062146.$$

The failure probabilities rise to $(0.013)(0.01)/0.50 = 0.00026$ and $(0.10)(0.01)/0.50 = 0.002$, and the true social costs to $S = (0.005)(0.082548) + (0.50)(0.00026) = 0.00041274 + 0.00013 = 0.00054274$ and $S = (0.005)(0.062146) + (0.50)(0.002) = 0.00031073 + 0.001 = 0.00131073$, both above $S(e^*) = 0.00051052$.

Answer (d). (i) 8.25%, (ii) 6.21%, both below $e^* = 9.21\%$; the larger the cost wrongly assigned to equity, the thinner the "optimal" capital and the higher the true social cost.

Step 7 — (e) Comparative statics. Differentiate (18.4), written as $e^* = \theta\big[\ln\kappa - \ln\lambda - \ln\theta\big]$:

$$\frac{\partial e^*}{\partial \kappa} = \frac{\theta}{\kappa} = \frac{0.01}{0.50} = 0.02 > 0,$$
$$\frac{\partial e^*}{\partial \lambda} = -\frac{\theta}{\lambda} = -\frac{0.01}{0.005} = -2 < 0,$$
$$\frac{\partial e^*}{\partial \theta} = \big[\ln\kappa - \ln\lambda - \ln\theta\big] + \theta\Big(-\frac{1}{\theta}\Big) = \ln\frac{\kappa}{\lambda\theta} - 1 = 9.2103 - 1 = 8.2103 > 0.$$

Doubling κ changes e^* by

$$\theta\ln\frac{2\kappa}{\lambda\theta} - \theta\ln\frac{\kappa}{\lambda\theta} = \theta\ln 2 = (0.01)(0.6931) = 0.0069315,$$

from 9.21% to $(0.01)\ln(20{,}000) = 9.90\%$.

Answer (e). e^* rises with κ (+0.02 per unit) and with θ (+8.21 per unit, since $\kappa/(\lambda\theta) > e$), and falls with λ (-2 per unit). Doubling κ adds 0.69 percentage points.

Intuition. The bank's leverage choice is a trade-off in which it bears all of the benefit of debt and none of the cost of failure. Society's trade-off counts both, and once the MM "costs" of equity are seen to be risk compensation and transfers, the only real cost of equity left is the small loss of liquidity services. That is why the social optimum is well above what banks would choose, and why getting Statements 2 and 3 wrong biases capital requirements downward.

Common mistakes.

- Putting the equity premium or the tax shield into the social cost; both drop out by MM.
- Stating the first-order condition without differentiating $\exp(-e/\theta)$ (the factor $-1/\theta$ is where θ enters e^*).
- Reading the small private cost in (c) as proof that capital is cheap for society, or the reverse: the private cost includes a transfer, the social gain is the externality avoided.

Ref: Prof. Ricca, Topic 2B slides 44–45 (static trade-off: marginal benefit of debt = marginal expected bankruptcy cost, $V_L = V_U + V_{ITS} - PV$(bankruptcy costs)), slides 46–47 (bank capital debate, failure externality).